

- (c) Conclude from (b) that the number of linearly independent columns of A is the same as the number of linearly independent columns of B .
23. Let A' be a matrix in reduced row echelon form.
- (a) Prove that the nonzero rows of A' are linearly independent. Prove that the pivotal columns of A' are linearly independent and that the nonpivotal columns of A' are linearly dependent on the pivotal columns. (Note the role the pivotal elements play.)
- (b) Prove that the number of linearly independent columns of a matrix in reduced row echelon form is the same as the number of linearly independent rows, i.e., the row rank and the column rank of such a matrix are the same.
24. Use the previous two exercises and Exercise 15 above to prove in general that the row rank and the column rank of a matrix are the same.
25. (*Computing Inverses of Matrices*) Let A be an $n \times n$ matrix.
- (a) Show that A has an inverse matrix B with columns $B_1, B_2, \dots, B_n$ if and only if the systems of equations:

$$AB_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, \quad AB_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, \quad \dots, \quad AB_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

- have solutions.
- (b) Prove that A has an inverse if and only if A is row equivalent to the $n \times n$ identity matrix.
- (c) Prove that A has an inverse B if and only if the augmented matrix $(A \mid I)$ can be row reduced to the augmented matrix $(I \mid B)$ where I is the $n \times n$ identity matrix.
26. Determine the inverses of the following matrices using row reduction:

$$A = \begin{pmatrix} -7 & -1 & -4 \\ 7 & 1 & 3 \\ 1 & 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 2 & 1 & -1 \\ 0 & 2 & 0 & 0 \\ -1 & 1 & 1 & 0 \end{pmatrix}.$$

27. (*Computing Spans, Linear Independence and Linear Dependencies in Vector Spaces*) Let V be an m -dimensional vector space with basis $e_1, e_2, \dots, e_m$ and let $v_1, v_2, \dots, v_n$ be vectors in V . Let A be the $m \times n$ matrix whose columns are the coordinates of the vectors v_i (with respect to the basis $e_1, e_2, \dots, e_m$) and let A' be the reduced row echelon form of A .
- (a) Let B be any matrix row equivalent to A . Let $w_1, w_2, \dots, w_n$ be the vectors whose coordinates (with respect to the basis $e_1, e_2, \dots, e_m$) are the columns of B . Prove that any linear relation
- $$x_1 v_1 + x_2 v_2 + \dots + x_n v_n = 0 \quad (11.5)$$
- satisfied by $v_1, v_2, \dots, v_n$ is also satisfied when v_i is replaced by $w_i, i = 1, 2, \dots, n$.
- (b) Prove that the vectors whose coordinates are given by the pivotal columns of A' are linearly independent and that the vectors whose coordinates are given by the nonpivotal columns of A' are linearly dependent on these.
- (c) (*Determining Linear Independence of Vectors*) Prove that the vectors $v_1, v_2, \dots, v_n$ are linearly independent if and only if A' has n nonzero rows (i.e., has rank n).
- (d) (*Determining Linear Dependencies of Vectors*) By (c), the vectors $v_1, v_2, \dots, v_n$ are linearly dependent if and only if A' has nonpivotal columns. The solutions to (5)

defining linear dependence relations among $v_1, v_2, \dots, v_n$ are given by the linear equations defined by A' . Show that each of the variables $x_1, x_2, \dots, x_n$ in (5) corresponding to the nonpivotal columns of A' can be prescribed arbitrarily and the values of the remaining variables are then uniquely determined to give a linear dependence relation among $v_1, v_2, \dots, v_n$ as in (5).

- (e) (*Determining the Span of a Set of Vectors*) Prove that the subspace W spanned by $v_1, v_2, \dots, v_n$ has dimension r where r is the number of nonzero rows of A' and that a basis for W is given by the original vectors v_{j_i} ($i = 1, 2, \dots, r$) corresponding to the pivotal columns of A' .

28. Let $V = \mathbb{R}^5$ with the standard basis and consider the vectors

$$v_1 = (1, 1, 3, -2, 3), \quad v_2 = (0, 1, 0, -1, 0), \quad v_3 = (2, 3, 6, -5, 6)$$

$$v_4 = (0, 3, 1, -3, 1), \quad v_5 = (2, -1, -1, -1, -1).$$

- (a) Show that the reduced row echelon form of the matrix

$$A = \begin{pmatrix} 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & 3 & 3 & -1 \\ 3 & 0 & 6 & 1 & -1 \\ -2 & -1 & -5 & -3 & -1 \\ 3 & 0 & 6 & 1 & -1 \end{pmatrix}$$

whose columns are the coordinates of v_1, v_2, v_3, v_4, v_5 is the matrix

$$A' = \begin{pmatrix} 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 0 & 18 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

where the 1st, 2nd and 4th columns are pivotal and the remaining two are nonpivotal.

- (b) Conclude that these vectors are linearly dependent, that the subspace W spanned by v_1, v_2, v_3, v_4, v_5 is 3-dimensional and that the vectors

$$v_1 = (1, 1, 3, -2, 3), \quad v_2 = (0, 1, 0, -1, 0) \quad \text{and} \quad v_4 = (0, 3, 1, -3, 1)$$

are a basis for W .

- (c) Conclude from (a) that the coefficients x_1, x_2, x_3, x_4, x_5 of any linear relation

$$x_1 v_1 + x_2 v_2 + x_3 v_3 + x_4 v_4 + x_5 v_5 = 0$$

satisfied by v_1, v_2, v_3, v_4, v_5 are given by the equations

$$\begin{aligned} x_1 + 2x_3 + 2x_5 &= 0 \\ x_2 + x_3 + 18x_5 &= 0 \\ x_4 - 7x_5 &= 0. \end{aligned}$$

Deduce that the 3rd and 5th variables, namely x_3 and x_5 , corresponding to the nonpivotal columns of A' , can be prescribed arbitrarily and the remaining variables are then uniquely determined as:

$$\begin{aligned} x_1 &= -2x_3 - 2x_5 \\ x_2 &= -x_3 - 18x_5 \\ x_4 &= 7x_5 \end{aligned}$$

to give all the linear dependence relations satisfied by v_1, v_2, v_3, v_4, v_5 . In particular show that

$$-2v_1 - v_2 + v_3 = 0$$

and

$$-2v_1 - 18v_2 + 7v_4 + v_5 = 0$$

corresponding to $(x_3 = 1, x_5 = 0)$ and $(x_3 = 0, x_5 = 1)$, respectively.

29. For each exercise below, determine whether the given vectors in $\mathbb{R}^4$ are linearly independent. If they are linearly dependent, determine an explicit linear dependence among them.

(a) $(1, -4, 3, 0), (0, -1, 4, -3), (1, -1, 1, -1), (2, 2, -1, -3)$.

(b) $(1, -2, 4, 1), (2, -3, 9, -1), (1, 0, 6, -5), (2, -5, 7, 5)$.

(c) $(1, -2, 0, 1), (2, -2, 0, 0), (-1, 3, 0, -2), (-2, 1, 0, 1)$.

(d) $(0, 1, 1, 0), (1, 0, 1, 1), (2, 2, 2, 0), (0, -1, 1, 1)$.

30. For each exercise below, determine the subspace spanned in $\mathbb{R}^4$ by the given vectors and give a basis for this subspace.

(a) $(1, -2, 5, 3), (2, 3, 1, -4), (3, 8, -3, -5)$.

(b) $(2, -5, 3, 0), (0, -2, 5, -3), (1, -1, 1, -1), (-3, 2, -1, 2)$.

(c) $(1, -2, 0, 1), (2, -2, 0, 0), (-1, 3, 0, -2), (-2, 1, 0, 1)$.

(d) $(1, 1, 0, -1), (1, 2, 3, 0), (2, 3, 3, -1), (1, 2, 2, -2), (2, 3, 2, -3), (1, 3, 4, -3)$.

31. (*Computing the Image and Kernel of a Linear Transformation*) Let V be an n -dimensional vector space with basis $e_1, e_2, \dots, e_n$ and let W be an m -dimensional vector space with basis $f_1, f_2, \dots, f_m$. Let φ be a linear transformation from V to W and let A be the corresponding $m \times n$ matrix with respect to these bases: $A = (a_{ij})$ where

$$\varphi(e_j) = \sum_{i=1}^m a_{ij} f_i, \quad j = 1, 2, \dots, n,$$

i.e., the columns of A are the coordinates of the vectors $\varphi(e_1), \varphi(e_2), \dots, \varphi(e_n)$ with respect to the basis $f_1, f_2, \dots, f_m$ of W . Let A' be the reduced row echelon form of A .

- (a) (*Determining the Image of a Linear Transformation*) Prove that the image $\varphi(V)$ of V under φ has dimension r where r is the number of nonzero rows of A' and that a basis for $\varphi(V)$ is given by the vectors $\varphi(e_{j_i})$ ($i = 1, 2, \dots, r$), i.e., the columns of A corresponding to the pivotal columns of A' give the coordinates of a basis for the image of φ .

- (b) (*Determining the Kernel of a Linear Transformation*) The elements in the kernel of φ are the vectors in V whose coordinates $(x_1, x_2, \dots, x_n)$ with respect to the basis $e_1, e_2, \dots, e_n$ satisfy the equation

$$A \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = 0,$$

and the solutions $x_1, x_2, \dots, x_n$ to this system of linear equations are determined by the matrix A' .

- (i) Prove that φ is injective if and only if A' has n nonzero rows (i.e., has rank n).
 (ii) By (i), the kernel of φ is nontrivial if and only if A' has nonpivotal columns. Show that each of the variables $x_1, x_2, \dots, x_n$ above corresponding to the nonpivotal columns of A' can be prescribed arbitrarily and the values of the remaining variables are then

uniquely determined to give an element $x_1e_1 + x_2e_2 + \dots + x_ne_n$ in the kernel of φ . In particular, show that the coordinates of a basis for the kernel are obtained by successively setting one nonpivotal variable equal to 1 and all other nonpivotal variables to 0 and solving for the remaining pivotal variables. Conclude that the kernel of φ has dimension $n - r$ where r is the rank of A .

32. Let $V = \mathbb{R}^5$ and $W = \mathbb{R}^4$ with the standard bases. Let φ be the linear transformation $\varphi : V \rightarrow W$ defined by

$$\varphi(x, y, z, u, v) = (x + 2y + 3z + 4u + 4v, -2x - 4y + 2v, x + 2y + u - 2v, x + 2y - v).$$

- (a) Prove that the matrix A corresponding to φ and these bases is

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 & 4 \\ -2 & -4 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & -2 \\ 1 & 2 & 0 & 0 & -1 \end{pmatrix}$$

and that the reduced row echelon matrix A' row equivalent to A is

$$A' = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

where the 1st, 3rd and 4th columns are pivotal and the remaining two are nonpivotal.

- (b) Conclude that the image of φ is 3-dimensional and that the image of the 1st, 3rd and 4th basis elements of V , namely, $(1, -2, 1, 1)$, $(3, 0, 0, 0)$ and $(4, 0, 1, 0)$ give a basis for the image $\varphi(V)$ of V .
- (c) Conclude from (a) that the elements in the kernel of φ are the vectors (x, y, z, u, v) satisfying the equations

$$\begin{aligned} x + 2y - v &= 0 \\ z + 3v &= 0 \\ u - v &= 0. \end{aligned}$$

Deduce that the 2nd and 5th variables, namely y and v , corresponding to the nonpivotal columns of A' can be prescribed arbitrarily and the remaining variables are then uniquely determined as

$$\begin{aligned} x &= -2y + v \\ z &= -3v \\ u &= v. \end{aligned}$$

Show that $(-2, 1, 0, 0, 0)$ and $(1, 0, -3, 1, 1)$ give a basis for the 2-dimensional kernel of φ , corresponding to $(y = 1, v = 0)$ and $(y = 0, v = 1)$, respectively.

33. Let φ be the linear transformation from $\mathbb{R}^4$ to itself defined by the matrix

$$A = \begin{pmatrix} 1 & -1 & 0 & 3 \\ -1 & 2 & 1 & -1 \\ -1 & 1 & 0 & -3 \\ 1 & -2 & -1 & 1 \end{pmatrix}$$

with respect to the standard basis for $\mathbb{R}^4$. Determine a basis for the image and for the kernel of φ .

34. Let φ be the linear transformation $\varphi : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ such that

$$\begin{aligned}\varphi((1, 0, 0, 0)) &= (1, -1) & \varphi((1, -1, 0, 0)) &= (0, 0) \\ \varphi((1, -1, 1, 0)) &= (1, -1) & \varphi((1, -1, 1, -1)) &= (0, 0).\end{aligned}$$

Determine a basis for the image and for the kernel of φ .

35. Let V be the set of all 2×2 matrices with real entries and let $\varphi : V \rightarrow \mathbb{R}$ be the map defined by sending a matrix $A \in V$ to the sum of the diagonal entries of A (the *trace* of A).

- (a) Show that

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

is a basis for V .

- (b) Prove that φ is a linear transformation and determine the matrix of φ with respect to the basis in (a) for V . Determine the dimension of and a basis for the kernel of φ .
36. Let V be the 6-dimensional vector space over $\mathbb{Q}$ consisting of the polynomials in the variable x of degree at most 5. Let φ be the map of V to itself defined by $\varphi(f) = x^2 f'' - 6x f' + 12f$, where f'' denotes the usual second derivative (with respect to x) of the polynomial $f \in V$ and f' similarly denotes the usual first derivative.
- (a) Prove that φ is a linear transformation of V to itself.
- (b) Determine a basis for the image and for the kernel of φ .
37. Let V be the 7-dimensional vector space over the field F consisting of the polynomials in the variable x of degree at most 6. Let φ be the linear transformation of V to itself defined by $\varphi(f) = f'$, where f' denotes the usual derivative (with respect to x) of the polynomial $f \in V$. For each of the fields below, determine a basis for the image and for the kernel of φ :
- (a) $F = \mathbb{R}$
- (b) $F = \mathbb{F}_2$, the finite field of 2 elements (note that, for example, $(x^2)' = 2x = 0$ over this field)
- (c) $F = \mathbb{F}_3$
- (d) $F = \mathbb{F}_5$.
38. Let A and B be square matrices. Prove that the trace of their Kronecker product is the product of their traces: $\text{tr}(A \otimes B) = \text{tr}(A) \text{tr}(B)$. (Recall that the trace of a square matrix is the sum of its diagonal entries.)
39. Let F be a subfield of K and let $\psi : V \rightarrow W$ be a linear transformation of finite dimensional vector spaces over F .
- (a) Prove that $1 \otimes \psi$ is a K -linear transformation from the vector spaces $K \otimes_F V$ to $K \otimes_F W$ over K . (Here 1 denotes the identity map from K to itself.)
- (b) Let $\mathcal{B} = \{v_1, \dots, v_n\}$ and $\mathcal{E} = \{w_1, \dots, w_m\}$ be bases of V and W respectively. Prove that the matrix of $1 \otimes \psi$ with respect to the bases $\{1 \otimes v_1, \dots, 1 \otimes v_n\}$ and $\{1 \otimes w_1, \dots, 1 \otimes w_m\}$ is the same as the matrix of ψ with respect to $\mathcal{B}$ and $\mathcal{E}$.

11.3 DUAL VECTOR SPACES

Definition.

- (1) For V any vector space over F let $V^* = \text{Hom}_F(V, F)$ be the space of linear transformations from V to F , called the *dual space* of V . Elements of V^* are called *linear functionals*.

(2) If $B = \{v_1, v_2, \dots, v_n\}$ is a basis of the finite dimensional space V , define $v_i^* \in V^*$ for each $i \in \{1, 2, \dots, n\}$ by its action on the basis B :

$$v_i^*(v_j) = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases} \quad 1 \leq j \leq n. \quad (11.6)$$

Proposition 18. With notations as above, $\{v_1^*, v_2^*, \dots, v_n^*\}$ is a basis of V^* . In particular, if V is finite dimensional then V^* has the same dimension as V .

Proof: Observe that since V is finite dimensional, $\dim V^* = \dim \text{Hom}_F(V, F) = \dim V = n$ (Corollary 11), so since there are n of the v_i^* 's it suffices to prove that they are linearly independent. If

$$\alpha_1 v_1^* + \alpha_2 v_2^* + \dots + \alpha_n v_n^* = 0 \quad \text{in } \text{Hom}_F(V, F),$$

then applying this element to v_i and using equation (6) above we obtain $\alpha_i = 0$. Since i is arbitrary these elements are linearly independent.

Definition. The basis $\{v_1^*, v_2^*, \dots, v_n^*\}$ of V^* is called the *dual basis* to $\{v_1, v_2, \dots, v_n\}$.

The exercises later show that if V is infinite dimensional it is always true that $\dim V < \dim V^*$. For spaces of arbitrary dimension the space V^* is the “algebraic” dual space to V . If V has some additional structure, for example a continuous structure (i.e., a topology), then one may define other types of dual spaces (e.g., the continuous dual of V , defined by requiring the linear functionals to be *continuous* maps). One has to be careful when reading other works (particularly analysis books) to ascertain what qualifiers are implicit in the use of the terms “dual space” and “linear functional.”

Example

Let $[a, b]$ be a closed interval in $\mathbb{R}$ and let V be the real vector space of all continuous functions $f : [a, b] \rightarrow \mathbb{R}$. If $a < b$, V is infinite dimensional. For each $g \in V$ the function $\varphi_g : V \rightarrow \mathbb{R}$ defined by $\varphi_g(f) = \int_a^b f(t)g(t)dt$ is a linear functional on V .

Definition. The dual of V^* , namely V^{**} , is called the *double dual* or *second dual* of V .

Note that for a finite dimensional space V , $\dim V = \dim V^*$ and also $\dim V^* = \dim V^{**}$, hence V and V^{**} are isomorphic vector spaces. For infinite dimensional spaces $\dim V < \dim V^{**}$ (cf. the exercises) so V and V^{**} cannot be isomorphic. In the case of finite dimensional spaces there is a *natural*, i.e., basis independent or coordinate free way of exhibiting the isomorphism between a vector space and its second dual. The basic idea, in a more general setting, is as follows: if X is any set and S is any set of functions of X into the field F , we normally think of choosing or fixing an $f \in S$ and computing $f(x)$ as x ranges over all of X . Alternatively, we could think of fixing a point x in X and computing $f(x)$ as f ranges over all of S . The latter process, called *evaluation at x* shows that for each $x \in X$ there is a function $E_x : S \rightarrow F$ defined by

$E_x(f) = f(x)$ (i.e., evaluate f at x). This gives a map $x \mapsto E_x$ of X into the set of F -valued functions on S . If S “separates points” in the sense that for distinct points x and y of X there is some $f \in S$ such that $f(x) \neq f(y)$, then the map $x \mapsto E_x$ is injective. The proof of the next lemma applies this “role reversal” process to the situation where $X = V$ and $S = V^*$, proves E_x is a linear F -valued function on S , that is, E_x belongs to the dual space of V^* , and proves the map $x \mapsto E_x$ is a linear transformation from V into V^{**} . Note that throughout this process there is no mention of the word “basis” (although it is convenient to know the dimension of V^{**} — a fact we established by picking bases). In particular, the proof does not start with the familiar phrase “pick a basis of $V \dots$.”

Theorem 19. There is a natural injective linear transformation from V to V^{**} . If V is finite dimensional then this linear transformation is an isomorphism.

Proof: Let $v \in V$. Define the map (*evaluation at v*)

$$E_v : V^* \rightarrow F \quad \text{by} \quad E_v(f) = f(v).$$

Then $E_v(f + \alpha g) = (f + \alpha g)(v) = f(v) + \alpha g(v) = E_v(f) + \alpha E_v(g)$, so that E_v is a linear transformation from V^* to F . Hence E_v is an element of $\text{Hom}_F(V^*, F) = V^{**}$. This defines a natural map

$$\varphi : V \rightarrow V^{**} \quad \text{by} \quad \varphi(v) = E_v.$$

The map φ is a *linear* map, as follows: for $v, w \in V$ and $\alpha \in F$,

$$E_{v+\alpha w}(f) = f(v + \alpha w) = f(v) + \alpha f(w) = E_v(f) + \alpha E_w(f)$$

for every $f \in V^*$, and so

$$\varphi(v + \alpha w) = E_{v+\alpha w} = E_v + \alpha E_w = \varphi(v) + \alpha \varphi(w).$$

To see that φ is injective let v be any nonzero vector in V . By the Building Up Lemma there is a basis $\mathcal{B}$ containing v . Let f be the linear transformation from V to F defined by sending v to 1 and every element of $\mathcal{B} - \{v\}$ to zero. Then $f \in V^*$ and $E_v(f) = f(v) = 1$. Thus $\varphi(v) = E_v$ is not zero in V^{**} . This proves $\ker \varphi = 0$, i.e., φ is injective.

If V has finite dimension n then by Proposition 18, V^* and hence also V^{**} has dimension n . In this case φ is an injective linear transformation from V to a finite dimensional vector space of the same dimension, hence is an isomorphism.

Let V, W be finite dimensional vector spaces over F with bases $\mathcal{B}, \mathcal{E}$, respectively and let $\mathcal{B}^*, \mathcal{E}^*$ be the dual bases. Fix some $\varphi \in \text{Hom}_F(V, W)$. Then for each $f \in W^*$, the composite $f \circ \varphi$ is a linear transformation from V to F , that is $f \circ \varphi \in V^*$. Thus the map $f \mapsto f \circ \varphi$ defines a function from W^* to V^* . We denote this induced function on dual spaces by φ^* .

Theorem 20. With notations as above, φ^* is a linear transformation from W^* to V^* and $M_{\mathcal{E}^*}^{\mathcal{B}^*}(\varphi^*)$ is the transpose of the matrix $M_{\mathcal{B}}^{\mathcal{E}}(\varphi)$ (recall that the transpose of the matrix (a_{ij}) is the matrix (a_{ji})).

Proof: The map φ^* is linear because $(f + \alpha g) \circ \varphi = (f \circ \varphi) + \alpha(g \circ \varphi)$. The equations which define φ are (from its matrix)

$$\varphi(v_j) = \sum_{i=1}^m \alpha_{ij} w_i \quad 1 \leq j \leq n.$$

To compute the matrix for φ^* , observe that by the definitions of φ^* and w_k^*

$$\varphi^*(w_k^*)(v_j) = (w_k^* \circ \varphi)(v_j) = w_k^* \left(\sum_{i=1}^m \alpha_{ij} w_i \right) = \alpha_{kj}.$$

Also

$$\left(\sum_{i=1}^n \alpha_{ki} v_i^* \right) (v_j) = \alpha_{kj}$$

for all j . This shows that the two linear functionals below agree on a basis of V , hence they are the same element of V^* :

$$\varphi^*(w_k^*) = \sum_{i=1}^n \alpha_{ki} v_i^*.$$

This determines the matrix for φ^* with respect to the bases $\mathcal{E}^*$ and $\mathcal{B}^*$ as the transpose of the matrix for φ .

Corollary 21. For any matrix A , the row rank of A equals the column rank of A .

Proof: Let $\varphi : V \rightarrow W$ be a linear transformation whose matrix with respect to some fixed bases of V and W is A . By Theorem 20 the matrix of $\varphi^* : W^* \rightarrow V^*$ with respect to the dual bases is the transpose of A . The column rank of A is the rank of φ and the row rank of A (= the column rank of the transpose of A) is the rank of φ^* (cf. Exercise 6 of Section 2). It therefore suffices to show that φ and φ^* have the same rank. Now

$$\begin{aligned} f \in \ker \varphi^* &\Leftrightarrow \varphi^*(f) = 0 \Leftrightarrow f \circ \varphi(v) = 0, \quad \text{for all } v \in V \\ &\Leftrightarrow \varphi(V) \subseteq \ker f \Leftrightarrow f \in \text{Ann}(\varphi(V)), \end{aligned}$$

where $\text{Ann}(S)$ is the annihilator of S described in Exercise 3 below. Thus $\text{Ann}(\varphi(V)) = \ker \varphi^*$. By Exercise 3, $\dim \text{Ann}(\varphi(V)) = \dim W^* - \dim \varphi(V)$. By Corollary 8, $\dim \ker \varphi^* = \dim W^* - \dim \varphi^*(W^*)$. Since W and W^* have the same dimension, $\dim \varphi(V) = \dim \varphi^*(W^*)$ as needed.

EXERCISES

1. Let V be a finite dimensional vector space. Prove that the map $\varphi \mapsto \varphi^*$ in Theorem 20 gives a ring isomorphism of $\text{End}(V)$ with $\text{End}(V^*)$.
2. Let V be the collection of polynomials with coefficients in $\mathbb{Q}$ in the variable x of degree at most 5 with $1, x, x^2, \dots, x^5$ as basis. Prove that the following are elements of the dual space of V and express them as linear combinations of the dual basis:
 - (a) $E : V \rightarrow \mathbb{Q}$ defined by $E(p(x)) = p(3)$ (i.e., evaluation at $x = 3$).
 - (b) $\varphi : V \rightarrow \mathbb{Q}$ defined by $\varphi(p(x)) = \int_0^1 p(t) dt$.
 - (c) $\varphi : V \rightarrow \mathbb{Q}$ defined by $\varphi(p(x)) = \int_0^1 t^2 p(t) dt$.
 - (d) $\varphi : V \rightarrow \mathbb{Q}$ defined by $\varphi(p(x)) = p'(5)$ where $p'(x)$ denotes the usual derivative of the polynomial $p(x)$ with respect to x .
3. Let S be any subset of V^* for some finite dimensional space V . Define $\text{Ann}(S) = \{v \in V \mid f(v) = 0 \text{ for all } f \in S\}$. ($\text{Ann}(S)$ is called the *annihilator of S in V*).
- (a) Prove that $\text{Ann}(S)$ is a subspace of V .
- (b) Let W_1 and W_2 be subspaces of V^* . Prove that $\text{Ann}(W_1 + W_2) = \text{Ann}(W_1) \cap \text{Ann}(W_2)$ and $\text{Ann}(W_1 \cap W_2) = \text{Ann}(W_1) + \text{Ann}(W_2)$.
- (c) Let W_1 and W_2 be subspaces of V^* . Prove that $W_1 = W_2$ if and only if $\text{Ann}(W_1) = \text{Ann}(W_2)$.
- (d) Prove that the annihilator of S is the same as the annihilator of the subspace of V^* spanned by S .
- (e) Assume V is finite dimensional with basis $v_1, \dots, v_n$. Prove that if $S = \{v_1^*, \dots, v_k^*\}$ for some $k \leq n$, then $\text{Ann}(S)$ is the subspace spanned by $\{v_{k+1}, \dots, v_n\}$.
- (f) Assume V is finite dimensional. Prove that if W^* is any subspace of V^* then $\dim \text{Ann}(W^*) = \dim V - \dim W^*$.
4. If V is infinite dimensional with basis $\mathcal{A}$, prove that $\mathcal{A}^* = \{v^* \mid v \in \mathcal{A}\}$ does *not* span V^* .
5. If V is infinite dimensional with basis $\mathcal{A}$, prove that V^* is isomorphic to the direct product of copies of F indexed by $\mathcal{A}$. Deduce that $\dim V^* > \dim V$. [Use Exercise 14, Section 1.]

11.4 DETERMINANTS

Although we shall be using the theory primarily for vector spaces over a field, the theory of determinants can be developed with no extra effort over arbitrary commutative rings with 1. Thus in this section R is any commutative ring with 1 and $V_1, V_2, \dots, V_n, V$ and W are R -modules. For convenience we repeat the definition of multilinear functions from Section 10.4.

Definition.

- (1) A map $\varphi : V_1 \times V_2 \times \dots \times V_n \rightarrow W$ is called *multilinear* if for each fixed i and fixed elements $v_j \in V_j, j \neq i$, the map

$$V_i \rightarrow W \quad \text{defined by} \quad x \mapsto \varphi(v_1, \dots, v_{i-1}, x, v_{i+1}, \dots, v_n)$$

is an R -module homomorphism. If $V_i = V, i = 1, 2, \dots, n$, then φ is called an *n -multilinear function on V* , and if in addition $W = R$, φ is called an *n -multilinear form on V* .

- (2) An n -multilinear function φ on V is called *alternating* if $\varphi(v_1, v_2, \dots, v_n) = 0$ whenever $v_i = v_{i+1}$ for some $i \in \{1, 2, \dots, n-1\}$ (i.e., φ is zero whenever two consecutive arguments are equal). The function φ is called *symmetric* if interchanging v_i and v_j for any i and j in $(v_1, v_2, \dots, v_n)$ does not alter the value of φ on this n -tuple.

When $n = 2$ (respectively, 3) one says φ is *bilinear* (respectively, *trilinear*) rather than 2-multilinear (respectively, 3-multilinear). Also, when n is clear from the context we shall simply say φ is multilinear.

Example

For any fixed $m \geq 0$ the usual dot product on $V = \mathbb{R}^m$ is a bilinear form (here the ring R is the field of real numbers).

Proposition 22. Let φ be an n -multilinear alternating function on V . Then

- (1) $\varphi(v_1, \dots, v_{i-1}, v_{i+1}, v_i, v_{i+2}, \dots, v_n) = -\varphi(v_1, v_2, \dots, v_n)$ for any $i \in \{1, 2, \dots, n-1\}$, i.e., the value of φ on an n -tuple is negated if two adjacent components are interchanged.
- (2) For each $\sigma \in S_n$, $\varphi(v_{\sigma(1)}, v_{\sigma(2)}, \dots, v_{\sigma(n)}) = \epsilon(\sigma)\varphi(v_1, v_2, \dots, v_n)$, where $\epsilon(\sigma)$ is the sign of the permutation σ (cf. Section 3.5).
- (3) If $v_i = v_j$ for any pair of distinct $i, j \in \{1, 2, \dots, n\}$ then $\varphi(v_1, v_2, \dots, v_n) = 0$.
- (4) If v_i is replaced by $v_i + \alpha v_j$ in $(v_1, \dots, v_n)$ for any $j \neq i$ and any $\alpha \in R$, the value of φ on this n -tuple is not changed.

Proof: (1) Let $\psi(x, y)$ be the function φ with variable entries x and y in positions i and $i+1$ respectively and fixed entries v_j in position j , for all other j . Thus (1) is the same as showing $\psi(x, y) = -\psi(y, x)$. Since φ is alternating $\psi(x+y, x+y) = 0$. Expanding $x+y$ in each variable in turn gives $\psi(x+y, x+y) = \psi(x, x) + \psi(x, y) + \psi(y, x) + \psi(y, y)$. Again, by the alternating property of φ , the first and last terms on the right hand side of the latter equation are zero. Thus $0 = \psi(x, y) + \psi(y, x)$, which gives (1).

(2) Every permutation can be written as a product of transpositions (cf. Section 3.5). Furthermore, every transposition may be written as a product of transpositions which interchange two successive integers (cf. Exercise 3 of Section 3.5). Thus every permutation σ can be written as $\tau_1 \cdots \tau_m$, where τ_k is a transposition interchanging two successive integers, for all k . It follows from m applications of (1) that

$$\varphi(v_{\sigma(1)}, v_{\sigma(2)}, \dots, v_{\sigma(n)}) = \epsilon(\tau_m) \cdots \epsilon(\tau_1) \varphi(v_1, v_2, \dots, v_n).$$

Finally, since ϵ is a homomorphism into the abelian group ± 1 (so the order of the factors ± 1 does not matter), $\epsilon(\tau_1) \cdots \epsilon(\tau_m) = \epsilon(\tau_1 \cdots \tau_m) = \epsilon(\sigma)$. This proves (2).

(3) Choose σ to be any permutation which fixes i and moves j to $i+1$. Thus $(v_{\sigma(1)}, v_{\sigma(2)}, \dots, v_{\sigma(n)})$ has two equal adjacent components so φ is zero on this n -tuple. By (2), $\varphi(v_{\sigma(1)}, v_{\sigma(2)}, \dots, v_{\sigma(n)}) = \pm \varphi(v_1, v_2, \dots, v_n)$. This implies (3).

(4) This follows immediately from (3) on expanding by linearity in the i^{th} position.

Proposition 23. Assume φ is an n -multilinear alternating function on V and that for some $v_1, v_2, \dots, v_n$ and $w_1, w_2, \dots, w_n \in V$ and some $\alpha_{ij} \in R$ we have

$$\begin{aligned} w_1 &= \alpha_{11}v_1 + \alpha_{21}v_2 + \cdots + \alpha_{n1}v_n \\ w_2 &= \alpha_{12}v_1 + \alpha_{22}v_2 + \cdots + \alpha_{n2}v_n \\ &\vdots \\ w_n &= \alpha_{1n}v_1 + \alpha_{2n}v_2 + \cdots + \alpha_{nn}v_n \end{aligned}$$

(we have purposely written the indices of the α_{ij} in “column format”). Then

$$\varphi(w_1, w_2, \dots, w_n) = \sum_{\sigma \in S_n} \epsilon(\sigma) \alpha_{\sigma(1)1} \alpha_{\sigma(2)2} \cdots \alpha_{\sigma(n)n} \varphi(v_1, v_2, \dots, v_n).$$

Proof: If we expand $\varphi(w_1, w_2, \dots, w_n)$ by multilinearity we obtain a sum of n^n terms of the form $\alpha_{i_1 1} \alpha_{i_2 2} \cdots \alpha_{i_n n} \varphi(v_{i_1}, v_{i_2}, \dots, v_{i_n})$, where the indices $i_1, i_2, \dots, i_n$ each run over $1, 2, \dots, n$. By Proposition 22(3), φ is zero on the terms where two or more of the i_j 's are equal. Thus in this expansion we need only consider the terms where $i_1, \dots, i_n$ are distinct. Such sequences are in bijective correspondence with permutations in S_n , so each nonzero term may be written as $\alpha_{\sigma(1)1} \alpha_{\sigma(2)2} \cdots \alpha_{\sigma(n)n} \varphi(v_{\sigma(1)}, v_{\sigma(2)}, \dots, v_{\sigma(n)})$, for some $\sigma \in S_n$. Applying (2) of the previous proposition to each of these terms in the expansion of $\varphi(w_1, w_2, \dots, w_n)$ gives the expression in the proposition.

Definition. An $n \times n$ determinant function on R is any function

$$\det : M_{n \times n}(R) \rightarrow R$$

that satisfies the following two axioms:

- (1) $\det$ is an n -multilinear alternating form on $R^n (= V)$, where the n -tuples are the n columns of the matrices in $M_{n \times n}(R)$
- (2) $\det(I) = 1$, where I is the $n \times n$ identity matrix.

On occasion we shall write $\det(A_1, A_2, \dots, A_n)$ for $\det A$, where $A_1, A_2, \dots, A_n$ are the columns of A .

Theorem 24. There is a unique $n \times n$ determinant function on R and it can be computed for any $n \times n$ matrix (α_{ij}) by the formula:

$$\det(\alpha_{ij}) = \sum_{\sigma \in S_n} \epsilon(\sigma) \alpha_{\sigma(1)1} \alpha_{\sigma(2)2} \cdots \alpha_{\sigma(n)n}.$$

Proof: Let $A_1, A_2, \dots, A_n$ be the column vectors in a general $n \times n$ matrix (α_{ij}) . We leave it as an exercise to check that the formula given in the statement of the theorem does satisfy the axioms of a determinant function — this gives existence of a determinant

function. To prove uniqueness let e_i be the column n -tuple with 1 in position i and zeros in all other positions. Then

$$\begin{aligned} A_1 &= \alpha_{11}e_1 + \alpha_{21}e_2 + \cdots + \alpha_{n1}e_n \\ A_2 &= \alpha_{12}e_1 + \alpha_{22}e_2 + \cdots + \alpha_{n2}e_n \\ &\vdots \\ A_n &= \alpha_{1n}e_1 + \alpha_{2n}e_2 + \cdots + \alpha_{nn}e_n. \end{aligned}$$

By Proposition 23, $\det A = \sum_{\sigma \in S_n} \epsilon(\sigma) \alpha_{\sigma(1)1} \alpha_{\sigma(2)2} \cdots \alpha_{\sigma(n)n} \det(e_1, e_2, \dots, e_n)$. Since by axiom (2) of a determinant function $\det(e_1, e_2, \dots, e_n) = 1$, the value of $\det A$ is as claimed.

Corollary 25. The determinant is an n -multilinear function of the rows of $M_{n \times n}(R)$ and for any $n \times n$ matrix A , $\det A = \det(A^t)$, where A^t is the transpose of A .

Proof: The first statement is an immediate consequence of the second, so it suffices to prove that a matrix and its transpose have the same determinant. For $A = (\alpha_{ij})$ one calculates that

$$\det A^t = \sum_{\sigma \in S_n} \epsilon(\sigma) \alpha_{1\sigma(1)} \alpha_{2\sigma(2)} \cdots \alpha_{n\sigma(n)}.$$

Each number from 1 to n appears exactly once among $\sigma(1), \dots, \sigma(n)$ so we may rearrange the product $\alpha_{1\sigma(1)} \alpha_{2\sigma(2)} \cdots \alpha_{n\sigma(n)}$ as $\alpha_{\sigma^{-1}(1)1} \alpha_{\sigma^{-1}(2)2} \cdots \alpha_{\sigma^{-1}(n)n}$. Also, the homomorphism ϵ takes values in $\{\pm 1\}$ so $\epsilon(\sigma) = \epsilon(\sigma^{-1})$. Thus the sum for $\det A^t$ may be rewritten as

$$\sum_{\sigma \in S_n} \epsilon(\sigma^{-1}) \alpha_{\sigma^{-1}(1)1} \alpha_{\sigma^{-1}(2)2} \cdots \alpha_{\sigma^{-1}(n)n}.$$

The latter sum is over all permutations, so the index σ^{-1} may be replaced by σ . The resulting expression is the sum for $\det A$. This completes the proof.

Theorem 26. (Cramer's Rule) If $A_1, A_2, \dots, A_n$ are the columns of an $n \times n$ matrix A and $B = \beta_1 A_1 + \beta_2 A_2 + \cdots + \beta_n A_n$, for some $\beta_1, \dots, \beta_n \in R$, then

$$\beta_i \det A = \det(A_1, \dots, A_{i-1}, B, A_{i+1}, \dots, A_n).$$

Proof: This follows immediately from Proposition 22(3) on replacing the given expression for B in the i^{th} position and expanding by multilinearity in that position.

Corollary 27. If R is an integral domain, then $\det A = 0$ for $A \in M_n(R)$ if and only if the columns of A are R -linearly dependent as elements of the free R -module of rank n . Also, $\det A = 0$ if and only if the rows of A are R -linearly dependent.

Proof: Since $\det A = \det A^t$ the first sentence implies the second. Assume first that the columns of A are linearly dependent and

$$0 = \beta_1 A_1 + \beta_2 A_2 + \cdots + \beta_n A_n$$

is a dependence relation on the columns of A with, say, $\beta_i \neq 0$. By Cramer's Rule, $\beta_i \det A = 0$. Since R is an integral domain and $\beta_i \neq 0$, $\det A = 0$.

Conversely, assume the columns of A are independent. Consider the integral domain R as embedded in its quotient field F so that $M_{n \times n}(R)$ may be considered as a subring of $M_{n \times n}(F)$ (and note that the determinant function on the subring is the restriction of the determinant function from $M_{n \times n}(F)$). The columns of A in this way become elements of F^n . Any nonzero F -linear combination of the columns of A which is zero in F^n gives, by multiplying the coefficients by a common denominator, a nonzero R -linear dependence relation. The columns of A must therefore be independent vectors in F^n . Since A has n columns, these form a basis of F^n . Thus there are elements β_{ij} of F such that for each i , the i^{th} basis vector e_i in F^n may be expressed as

$$e_i = \beta_{1i}A_1 + \beta_{2i}A_2 + \cdots + \beta_{ni}A_n.$$

The $n \times n$ identity matrix is the one whose columns are $e_1, e_2, \dots, e_n$. By Proposition 23 (with $\varphi = \det$), the determinant of the identity matrix is some F -multiple of $\det A$. Since the determinant of the identity matrix is 1, $\det A$ cannot be zero. This completes the proof.

Theorem 28. For matrices $A, B \in M_{n \times n}(R)$, $\det AB = (\det A)(\det B)$.

Proof: Let $B = (\beta_{ij})$ and let $A_1, A_2, \dots, A_n$ be the columns of A . Then $C = AB$ is the $n \times n$ matrix whose j^{th} column is $C_j = \beta_{1j}A_1 + \beta_{2j}A_2 + \cdots + \beta_{nj}A_n$. By Proposition 23 applied to the multilinear function $\det$ we obtain

$$\det C = \det(C_1, \dots, C_n) = \left[\sum_{\sigma \in S_n} \epsilon(\sigma) \beta_{\sigma(1)1} \beta_{\sigma(2)2} \cdots \beta_{\sigma(n)n} \right] \det(A_1, \dots, A_n).$$

The sum inside the brackets is the formula for $\det B$, hence $\det C = (\det B)(\det A)$, as required (R is commutative).

Definition. Let $A = (\alpha_{ij})$ be an $n \times n$ matrix. For each i, j , let A_{ij} be the $(n-1) \times (n-1)$ matrix obtained from A by deleting its i^{th} row and j^{th} column (an $(n-1) \times (n-1)$ minor of A). Then $(-1)^{i+j} \det(A_{ij})$ is called the ij cofactor of A .

Theorem 29. (*The Cofactor Expansion Formula along the i^{th} row*) If $A = (\alpha_{ij})$ is an $n \times n$ matrix, then for each fixed $i \in \{1, 2, \dots, n\}$ the determinant of A can be computed from the formula

$$\det A = (-1)^{i+1} \alpha_{i1} \det A_{i1} + (-1)^{i+2} \alpha_{i2} \det A_{i2} + \cdots + (-1)^{i+n} \alpha_{in} \det A_{in}.$$

Proof: For each A let $D(A)$ be the element of R obtained from the cofactor expansion formula described above. We prove that D satisfies the axioms of a determinant function, hence is the determinant function. Proceed by induction on n . If $n = 1$, $D((\alpha)) = \alpha$, for all 1×1 matrices (α) and the result holds. Assume therefore that $n \geq 2$. To show that D is an alternating multilinear function of the columns, fix an index k and consider the k^{th} column as varying and all other columns as fixed. If $j \neq k$,

α_{ij} does not depend on k and $D(A_{ij})$ is linear in the k^{th} column by induction. Also, as the k^{th} column varies linearly so does α_{ik} , whereas $D(A_{ik})$ remains unchanged (the k^{th} column has been deleted from A_{ik}). Thus each term in the formula for D varies linearly in the k^{th} column. This proves D is multilinear in the columns.

To prove D is alternating assume columns k and $k+1$ of A are equal. If $j \neq k$ or $k+1$, the two equal columns of A become two equal columns in the matrix A_{ij} . By induction $D(A_{ij}) = 0$. The formula for D therefore has at most two nonzero terms: when $j = k$ and when $j = k+1$. The minor matrices A_{ik} and $A_{i,k+1}$ are identical and $\alpha_{ik} = \alpha_{i,k+1}$. Then the two remaining terms in the expansion for D , $(-1)^{i+k}\alpha_{ik}D(A_{ik})$ and $(-1)^{i+k+1}\alpha_{i,k+1}D(A_{i,k+1})$ are equal and appear with opposite signs, hence they cancel. Thus $D(A) = 0$ if A has two adjacent columns which are equal, i.e., D is alternating.

Finally, it follows easily from the formula and induction that $D(I) = 1$, where I is the identity matrix. This completes the induction.

Theorem 30. (*Cofactor Formula for the Inverse of a Matrix*) Let $A = (\alpha_{ij})$ be an $n \times n$ matrix and let B be the transpose of its matrix of cofactors, i.e., $B = (\beta_{ij})$, where $\beta_{ij} = (-1)^{i+j} \det A_{ji}$, $1 \leq i, j \leq n$. Then $AB = BA = (\det A)I$. Moreover, $\det A$ is a unit in R if and only if A is a unit in $M_{n \times n}(R)$; in this case the matrix $\frac{1}{\det A}B$ is the inverse of A .

Proof: The i, j entry of AB is $\alpha_{i1}\beta_{1j} + \alpha_{i2}\beta_{2j} + \cdots + \alpha_{in}\beta_{nj}$. By definition of the entries of B this equals

$$\alpha_{i1}(-1)^{j+1}D(A_{j1}) + \alpha_{i2}(-1)^{j+2}D(A_{j2}) + \cdots + \alpha_{in}(-1)^{j+n}D(A_{jn}). \quad (11.7)$$

If $i = j$, this is the cofactor expansion for $\det A$ along the i^{th} row. The diagonal entries of AB are thus all equal to $\det A$. If $i \neq j$, let $\bar{A}$ be the matrix A with the j^{th} row replaced by the i^{th} row, so $\det \bar{A} = 0$. By inspection $\bar{A}_{jk} = A_{jk}$ and $\alpha_{ik} = \bar{\alpha}_{jk}$ for every $k \in \{1, 2, \dots, n\}$. By making these substitutions in equation (7) for each $k = 1, 2, \dots, n$ one sees that the i, j entry in AB equals $\bar{\alpha}_{j1}(-1)^{1+j}D(\bar{A}_{j1}) + \cdots + \bar{\alpha}_{jn}(-1)^{n+j}D(\bar{A}_{jn})$. This expression is the cofactor expansion for $\det \bar{A}$ along the j^{th} row. Since, as noted above, $\det \bar{A} = 0$, this proves that all off diagonal terms of AB are zero, which proves that $AB = (\det A)I$.

It follows directly from the definition of B that the pair (A^t, B^t) satisfies the same hypotheses as the pair (A, B) . By what has already been shown it follows that $(BA)^t = A^t B^t = (\det A^t)I$. Since $\det A^t = \det A$ and the transpose of a diagonal matrix is itself, we obtain $BA = (\det A)I$ as well.

If $d = \det A$ is a unit in R , then $d^{-1}B$ is a matrix with entries in R whose product with A (on either side) is the identity, i.e., A is a unit in $M_{n \times n}(R)$. Conversely, assume that A is a unit in R with (2-sided) inverse matrix C . Since $\det C \in R$ and

$$1 = \det I = \det AC = (\det A)(\det C) = (\det C)(\det A),$$

it follows that $\det A$ has a 2-sided inverse in R , as needed. This completes all parts of the proof.

EXERCISES

1. Formulate and prove the cofactor expansion formula along the j^{th} column of a square matrix A .
2. Let F be a field and let $A_1, A_2, \dots, A_n$ be (column) vectors in F^n . Form the matrix A whose i^{th} column is A_i . Prove that these vectors form a basis of F^n if and only if $\det A \neq 0$.
3. Let R be any commutative ring with 1, let V be an R -module and let $x_1, x_2, \dots, x_n \in V$. Assume that for some $A \in M_{n \times n}(R)$,

$$A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0.$$

Prove that $(\det A)x_i = 0$, for all $i \in \{1, 2, \dots, n\}$.

4. (*Computing Determinants of Matrices*) This exercise outlines the use of Gauss–Jordan elimination (cf. the exercises in Section 2) to compute determinants. This is the most efficient general procedure for computing large determinants. Let A be an $n \times n$ matrix.
 - (a) Prove that the elementary row operations have the following effect on determinants:
 - (i) interchanging two rows changes the sign of the determinant
 - (ii) adding a multiple of one row to another does not alter the determinant
 - (iii) multiplying any row by a nonzero element u from F multiplies the determinant by u .
 - (b) Prove that $\det A$ is nonzero if and only if A is row equivalent to the $n \times n$ identity matrix. Suppose A can be row reduced to the identity matrix using a total of s row interchanges as in (i) and by multiplying rows by the nonzero elements $u_1, u_2, \dots, u_t$ as in (iii). Prove that $\det A = (-1)^s(u_1 u_2 \dots u_t)^{-1}$.
5. Compute the determinants of the following matrices using row reduction:

$$A = \begin{pmatrix} 5 & 4 & -6 \\ -2 & 0 & 2 \\ 3 & 4 & -2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 & -4 & 4 \\ 2 & -1 & 4 & -8 \\ 1 & 0 & 1 & -2 \\ 0 & 1 & -2 & 3 \end{pmatrix}.$$

6. (*Minkowski's Criterion*) Suppose A is an $n \times n$ matrix with real entries such that the diagonal elements are all positive, the off-diagonal elements are all negative and the row sums are all positive. Prove that $\det A \neq 0$. [Consider the corresponding system of equations $AX = 0$ and suppose there is a nontrivial solution $(x_1, \dots, x_n)$. If x_i has the largest absolute value show that the i^{th} equation leads to a contradiction.]

11.5 TENSOR ALGEBRAS, SYMMETRIC AND EXTERIOR ALGEBRAS

In this section R is any commutative ring with 1, and we assume the left and right actions of R on each R -module are the same. We shall primarily be interested in the special case when $R = F$ is a field, but the basic constructions hold in general.

Suppose M is an R -module. When tensor products were first introduced in Section 10.4 we spoke heuristically of forming “products” $m_1 m_2$ of elements of M , and we constructed a new module $M \otimes M$ generated by such “products” $m_1 \otimes m_2$. The “value” of this product is not in M , so this does not give a ring structure on M itself. If, however,

we iterate this by taking the “products” $m_1m_2m_3$ and $m_1m_2m_3m_4$, and all finite sums of such products, we can construct a ring containing M that is “universal” with respect to rings containing M (and, more generally, with respect to homomorphic images of M), as we now show.

For each integer $k \geq 1$, define

$$\mathcal{T}^k(M) = M \otimes_R M \otimes_R \cdots \otimes_R M \quad (k \text{ factors}),$$

and set $\mathcal{T}^0(M) = R$. The elements of $\mathcal{T}^k(M)$ are called k -tensors. Define

$$\mathcal{T}(M) = R \oplus \mathcal{T}^1(M) \oplus \mathcal{T}^2(M) \oplus \mathcal{T}^3(M) \cdots = \bigoplus_{k=0}^{\infty} \mathcal{T}^k(M).$$

Every element of $\mathcal{T}(M)$ is a finite linear combination of k -tensors for various $k \geq 0$. We identify M with $\mathcal{T}^1(M)$, so that M is an R -submodule of $\mathcal{T}(M)$.

Theorem 31. If M is any R -module over the commutative ring R then

(1) $\mathcal{T}(M)$ is an R -algebra containing M with multiplication defined by mapping

$$(m_1 \otimes \cdots \otimes m_i)(m'_1 \otimes \cdots \otimes m'_j) = m_1 \otimes \cdots \otimes m_i \otimes m'_1 \otimes \cdots \otimes m'_j$$

and extended to sums via the distributive laws. With respect to this multiplication $\mathcal{T}^i(M)\mathcal{T}^j(M) \subseteq \mathcal{T}^{i+j}(M)$.

(2) (*Universal Property*) If A is any R -algebra and $\varphi : M \rightarrow A$ is an R -module homomorphism, then there is a unique R -algebra homomorphism $\Phi : \mathcal{T}(M) \rightarrow A$ such that $\Phi|_M = \varphi$.

Proof: The map

$$\underbrace{M \times M \times \cdots \times M}_i \times \underbrace{M \times M \times \cdots \times M}_j \rightarrow \mathcal{T}^{i+j}(M)$$

defined by

$$(m_1, \dots, m_i, m'_1, \dots, m'_j) \mapsto m_1 \otimes \cdots \otimes m_i \otimes m'_1 \otimes \cdots \otimes m'_j$$

is R -multilinear, so induces a bilinear map $\mathcal{T}^i(M) \times \mathcal{T}^j(M)$ to $\mathcal{T}^{i+j}(M)$ which is easily checked to give a well defined multiplication satisfying (1) (cf. the proof of Proposition 21 in Section 10.4). To prove (2), assume that $\varphi : M \rightarrow A$ is an R -algebra homomorphism. Then

$$(m_1, m_2, \dots, m_k) \mapsto \varphi(m_1)\varphi(m_2) \cdots \varphi(m_k)$$

defines an R -multilinear map from $M \times \cdots \times M$ (k times) to A . This in turn induces a unique R -module homomorphism Φ from $\mathcal{T}^k(M)$ to A (Corollary 16 of Section 10.4) mapping $m_1 \otimes \cdots \otimes m_k$ to the element on the right hand side above. It is easy to check from the definition of the multiplication in (1) that the resulting uniquely defined map $\Phi : \mathcal{T}(M) \rightarrow A$ is an R -algebra homomorphism.

Definition. The ring $\mathcal{T}(M)$ is called the *tensor algebra* of M .

Proposition 32. Let V be a finite dimensional vector space over the field F with basis $\mathcal{B} = \{v_1, \dots, v_n\}$. Then the k -tensors

$$v_{i_1} \otimes v_{i_2} \otimes \cdots \otimes v_{i_k} \quad \text{with } v_{i_j} \in \mathcal{B}$$

are a vector space basis of $\mathcal{T}^k(V)$ over F (with the understanding that the basis vector is the element $1 \in F$ when $k = 0$). In particular, $\dim_F(\mathcal{T}^k(V)) = n^k$.

Proof: This follows immediately from Proposition 16 of Section 2.

Theorem 31 and Proposition 32 show that the space $\mathcal{T}(V)$ may be regarded as the *noncommutative polynomial algebra* over F in the (noncommuting) variables $v_1, \dots, v_n$. The analogous result also holds for finitely generated free modules over any commutative ring (using Corollary 19 in Section 10.4).

Examples

- (1) Let $R = \mathbb{Z}$ and let $M = \mathbb{Q}/\mathbb{Z}$. Then $(\mathbb{Q}/\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Q}/\mathbb{Z}) = 0$ (Example 4 following Corollary 12 in Section 10.4). Thus $\mathcal{T}(\mathbb{Q}/\mathbb{Z}) = \mathbb{Z} \oplus (\mathbb{Q}/\mathbb{Z})$, where addition is componentwise and the multiplication is given by $(r, \bar{p})(s, \bar{q}) = (rs, \overline{rq + sp})$. The ring $R/(x)$ of Exercise 4(d) in Section 9.3 is isomorphic to $\mathcal{T}(\mathbb{Q}/\mathbb{Z})$.
- (2) Let $R = \mathbb{Z}$ and let $M = \mathbb{Z}/n\mathbb{Z}$. Then $(\mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ (Example 3 following Corollary 12 in Section 10.4). Thus $\mathcal{T}^i(M) \cong M$ for all $i > 0$ and so $\mathcal{T}(\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z} \oplus (\mathbb{Z}/n\mathbb{Z}) \oplus (\mathbb{Z}/n\mathbb{Z}) \cdots$. It follows easily that $\mathcal{T}(\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}[x]/(nx)$.

Since $\mathcal{T}^i(M)\mathcal{T}^j(M) \subseteq \mathcal{T}^{i+j}(M)$, the tensor algebra $\mathcal{T}(M)$ has a natural “grading” or “degree” structure reminiscent of a polynomial ring.

Definition.

- (1) A ring S is called a *graded ring* if it is the direct sum of additive subgroups: $S = S_0 \oplus S_1 \oplus S_2 \oplus \cdots$ such that $S_i S_j \subseteq S_{i+j}$ for all $i, j \geq 0$. The elements of S_k are said to be *homogeneous of degree k* , and S_k is called the *homogeneous component of S of degree k* .
- (2) An ideal I of the graded ring S is called a *graded ideal* if $I = \bigoplus_{k=0}^{\infty} (I \cap S_k)$.
- (3) A ring homomorphism $\varphi : S \rightarrow T$ between two graded rings is called a *homomorphism of graded rings* if it respects the grading structures on S and T , i.e., if $\varphi(S_k) \subseteq T_k$ for $k = 0, 1, 2, \dots$.

Note that $S_0 S_0 \subseteq S_0$, which implies that S_0 is a subring of the graded ring S and then S is an S_0 -module. If S_0 is in the center of S and it contains an identity of S , then S is an S_0 -algebra. Note also that the ideal I is graded if whenever a sum $i_{k_1} + \cdots + i_{k_n}$ of homogeneous elements with distinct degrees $k_1, \dots, k_n$ is in I then each of the individual summands $i_{k_1}, \dots, i_{k_n}$ is itself in I .

Example

The polynomial ring $S = R[x_1, x_2, \dots, x_n]$ in n variables over the commutative ring R is an example of a graded ring. Here $S_0 = R$ and the homogeneous component of degree k is the subgroup of all R -linear combinations of monomials of degree k .

The ideal I generated by $x_1, \dots, x_n$ is a graded ideal: every polynomial with zero constant term may be written uniquely as a sum of homogeneous polynomials of degree $k > 0$, and each of these has zero constant term hence lies in I . More generally, an ideal is a graded ideal if and only if it can be generated by homogeneous polynomials (cf. Exercise 17 in Section 9.1).

Not every ideal of a graded ring need be a graded ideal. For example in the graded ring $\mathbb{Z}[x]$ the principal ideal J generated by $1 + x$ is not graded: $1 + x \in J$ and $1 \notin J$ so $1 + x$ cannot be written as a sum of homogeneous polynomials each of which belongs to J .

The next result shows that quotients of graded rings by graded ideals are again graded rings.

Proposition 33. Let S be a graded ring, let I be a graded ideal in S and let $I_k = I \cap S_k$ for all $k \geq 0$. Then S/I is naturally a graded ring whose homogeneous component of degree k is isomorphic to S_k/I_k .

Proof: The map

$$\begin{aligned} S = \bigoplus_{k=0}^{\infty} S_k &\longrightarrow \bigoplus_{k=0}^{\infty} (S_k/I_k) \\ (\dots, s_k, \dots) &\longmapsto (\dots, s_k \bmod I_k, \dots) \end{aligned}$$

is surjective with kernel $I = \bigoplus_{k=0}^{\infty} I_k$ and defines an isomorphism of graded rings. The details are left for the exercises.

Symmetric Algebras

The first application of Proposition 33 is in the construction of a commutative quotient ring of $\mathcal{T}(M)$ through which R -module homomorphisms from M to any commutative R -algebra must factor. This gives an “abelianized” version of Theorem 31. The construction is analogous to forming the commutator quotient G/G' of a group (cf. Section 5.4).

Definition. The *symmetric algebra* of an R -module M is the R -algebra obtained by taking the quotient of the tensor algebra $\mathcal{T}(M)$ by the ideal $\mathcal{C}(M)$ generated by all elements of the form $m_1 \otimes m_2 - m_2 \otimes m_1$, for all $m_1, m_2 \in M$. The symmetric algebra $\mathcal{T}(M)/\mathcal{C}(M)$ is denoted by $\mathcal{S}(M)$.

The tensor algebra $\mathcal{T}(M)$ is generated as a ring by $R = \mathcal{T}^0(M)$ and $M = \mathcal{T}^1(M)$, and these elements commute in the quotient ring $\mathcal{S}(M)$ by definition. It follows that the symmetric algebra $\mathcal{S}(M)$ is a commutative ring. The ideal $\mathcal{C}(M)$ is generated by homogeneous tensors of degree 2 and it follows easily that $\mathcal{C}(M)$ is a graded ideal. Then by Proposition 33 the symmetric algebra is a graded ring whose homogeneous component of degree k is $\mathcal{S}^k(M) = \mathcal{T}^k(M)/\mathcal{C}^k(M)$. Since $\mathcal{C}(M)$ consists of k -tensors

with $k \geq 2$, we have $\mathcal{C}(M) \cap M = 0$ and so the image of $M = \mathcal{T}^1(M)$ in $S(M)$ is isomorphic to M . Identifying M with its image we see that $S^1(M) = M$ and the symmetric algebra contains M . In a similar way $S^0(M) = R$, so the symmetric algebra is also an R -algebra. The R -module $S^k(M)$ is called the k^{th} symmetric power of M .

The first part of the next theorem shows that the elements of the k^{th} symmetric power of M can be considered as finite sums of simple tensors $m_1 \otimes \cdots \otimes m_k$ where tensors with the order of the factors permuted are identified. Recall also from Section 4 that a k -multilinear map $\varphi : M \times \cdots \times M \rightarrow N$ is said to be *symmetric* if $\varphi(m_1, \dots, m_k) = \varphi(m_{\sigma(1)}, \dots, m_{\sigma(k)})$ for all permutations σ of $1, 2, \dots, k$. (The definition is the same for modules over any commutative ring R as for vector spaces.)

Theorem 34. Let M be an R -module over the commutative ring R and let $S(M)$ be its symmetric algebra.

- (1) The k^{th} symmetric power, $S^k(M)$, of M is equal to $M \otimes \cdots \otimes M$ (k factors) modulo the submodule generated by all elements of the form

$$(m_1 \otimes m_2 \otimes \cdots \otimes m_k) - (m_{\sigma(1)} \otimes m_{\sigma(2)} \otimes \cdots \otimes m_{\sigma(k)})$$

for all $m_i \in M$ and all permutations σ in the symmetric group S_k .

- (2) (*Universal Property for Symmetric Multilinear Maps*) If $\varphi : M \times \cdots \times M \rightarrow N$ is a symmetric k -multilinear map over R then there is a unique R -module homomorphism $\Phi : S^k(M) \rightarrow N$ such that $\varphi = \Phi \circ \iota$, where

$$\iota : M \times \cdots \times M \rightarrow S^k(M)$$

is the map defined by

$$\iota(m_1, \dots, m_k) = m_1 \otimes \cdots \otimes m_n \bmod \mathcal{C}(M).$$

- (3) (*Universal Property for maps to commutative R -algebras*) If A is any commutative R -algebra and $\varphi : M \rightarrow A$ is an R -module homomorphism, then there is a unique R -algebra homomorphism $\Phi : S(M) \rightarrow A$ such that $\Phi|_M = \varphi$.

Proof: The k -tensors $\mathcal{C}^k(M)$ in the ideal $\mathcal{C}(M)$ are finite sums of elements of the form

$$m_1 \otimes \cdots \otimes m_{i-1} \otimes (m_i \otimes m_{i+1} - m_{i+1} \otimes m_i) \otimes m_{i+2} \otimes \cdots \otimes m_k$$

with $m_1, \dots, m_k \in M$ (where $k \geq 2$ and $1 \leq i < k$). This product gives a difference of two k -tensors which are equal except that two entries (in positions i and $i + 1$) have been transposed, i.e., gives the element in (1) of the theorem corresponding to the transposition $(i \ i+1)$ in the symmetric group S_k . Conversely, since any permutation σ in S_k can be written as a product of such transpositions it is easy to see that every element in (1) can be written as a sum of elements of the form above. This gives (1).

The proofs of (2) and (3) are very similar to the proofs of the corresponding “asymmetric” results (Corollary 16 of Section 10.4 and Theorem 31) noting that $\mathcal{C}^k(M)$ is contained in the kernel of any symmetric map from $\mathcal{T}^k(M)$ to N by part (1).

Corollary 35. Let V be an n -dimensional vector space over the field F . Then $S(V)$ is isomorphic as a graded F -algebra to the ring of polynomials in n variables over F (i.e., the isomorphism is also a vector space isomorphism from $S^k(V)$ onto the space of all homogeneous polynomials of degree k). In particular, $\dim_F(S^k(V)) = \binom{k+n-1}{n-1}$.

Proof: Let $\mathcal{B} = \{v_1, \dots, v_n\}$ be a basis of V . By Proposition 32 there is a bijection between a basis of $\mathcal{T}^k(V)$ and the set $\mathcal{B}^k$ of ordered k -tuples of elements from $\mathcal{B}$. Define two k -tuples in $\mathcal{B}^k$ to be equivalent if there is some permutation of the entries of one that gives the other — this is easily seen to be an equivalence relation on $\mathcal{B}^k$. Let $S(\mathcal{B}^k)$ denote the corresponding set of equivalence classes. Any symmetric k -multilinear function from V^k to a vector space over F will be constant on all of the basis tensors whose corresponding k -tuples lie in the same equivalence class; conversely, any function from $S(\mathcal{B}^k)$ can be uniquely extended to a symmetric k -multilinear function on V^k . It follows that the vector space over F with basis $S(\mathcal{B}^k)$ satisfies the universal property of $S^k(V)$ in Theorem 34(2), hence is isomorphic to $S^k(V)$. Each equivalence class has a unique representative of the form $(v_1^{a_1}, v_2^{a_2}, \dots, v_n^{a_n})$, where $v_i^{a_i}$ denotes the sequence $v_i, v_i, \dots, v_i$ taken a_i times, each $a_i \geq 0$, and $a_1 + \dots + a_n = k$. Thus there is a bijection between the basis $S^k(\mathcal{B})$ and the set $x_1^{a_1} \dots x_n^{a_n}$ of monic monomials of degree k in the polynomial ring $F[x_1, \dots, x_n]$. This bijection extends to an isomorphism of graded F -algebras, proving the first part of the corollary. The computation of the dimension of $S^k(V)$ (i.e., the number of monic monomials of degree k) is left as an exercise.

Exterior Algebras

Recall from Section 4 that a multilinear map $\varphi : M \times \dots \times M \rightarrow N$ is called *alternating* if $\varphi(m_1, \dots, m_k) = 0$ whenever $m_i = m_{i+1}$ for some i . (The definition is the same for any R -module as for vector spaces.) We saw that the determinant map was alternating, and was uniquely determined by some additional constraints. We can apply Proposition 33 to construct an algebra through which alternating multilinear maps must factor in a manner similar to the construction of the symmetric algebra (through which symmetric multilinear maps factor).

Definition. The *exterior algebra* of an R -module M is the R -algebra obtained by taking the quotient of the tensor algebra $\mathcal{T}(M)$ by the ideal $\mathcal{A}(M)$ generated by all elements of the form $m \otimes m$, for $m \in M$. The exterior algebra $\mathcal{T}(M)/\mathcal{A}(M)$ is denoted by $\bigwedge(M)$ and the image of $m_1 \otimes m_2 \otimes \dots \otimes m_k$ in $\bigwedge(M)$ is denoted by $m_1 \wedge m_2 \wedge \dots \wedge m_k$.

As with the symmetric algebra, the ideal $\mathcal{A}(M)$ is generated by homogeneous elements hence is a graded ideal. By Proposition 33 the exterior algebra is graded, with k^{th} homogeneous component $\bigwedge^k(M) = \mathcal{T}^k(M)/\mathcal{A}^k(M)$. We can again identify R with $\bigwedge^0(M)$ and M with $\bigwedge^1(M)$ and so consider M as an R -submodule of the R -algebra $\bigwedge(M)$. The R -module $\bigwedge^k(M)$ is called the k^{th} *exterior power* of M .

The multiplication

$$(m_1 \wedge \dots \wedge m_i) \wedge (m'_1 \wedge \dots \wedge m'_j) = m_1 \wedge \dots \wedge m_i \wedge m'_1 \wedge \dots \wedge m'_j$$

in the exterior algebra is called the *wedge* (or *exterior*) *product*. By definition of the quotient, this multiplication is alternating in the sense that the product $m_1 \wedge \cdots \wedge m_k$ is 0 in $\bigwedge(M)$ if $m_i = m_{i+1}$ for any $1 \leq i < k$. Then

$$\begin{aligned} 0 &= (m + m') \wedge (m + m') \\ &= (m \wedge m) + (m \wedge m') + (m' \wedge m) + (m' \wedge m') \\ &= (m \wedge m') + (m' \wedge m) \end{aligned}$$

shows that the multiplication is also anticommutative on simple tensors:

$$m \wedge m' = -m' \wedge m \quad \text{for all } m, m' \in M.$$

This anticommutativity does not extend to arbitrary products, however, i.e., we need not have $ab = -ba$ for all $a, b \in \bigwedge(M)$ (cf. Exercise 4).

Theorem 36. Let M be an R -module over the commutative ring R and let $\bigwedge(M)$ be its exterior algebra.

- (1) The k^{th} exterior power, $\bigwedge^k(M)$, of M is equal to $M \otimes \cdots \otimes M$ (k factors) modulo the submodule generated by all elements of the form

$$m_1 \otimes m_2 \otimes \cdots \otimes m_k \quad \text{where } m_i = m_j \text{ for some } i \neq j.$$

In particular,

$$m_1 \wedge m_2 \wedge \cdots \wedge m_k = 0 \quad \text{if } m_i = m_j \text{ for some } i \neq j.$$

- (2) (*Universal Property for Alternating Multilinear Maps*) If $\varphi : M \times \cdots \times M \rightarrow N$ is an alternating k -multilinear map then there is a unique R -module homomorphism $\Phi : \bigwedge^k(M) \rightarrow N$ such that $\varphi = \Phi \circ \iota$, where

$$\iota : M \times \cdots \times M \rightarrow \bigwedge^k(M)$$

is the map defined by

$$\iota(m_1, \dots, m_k) = m_1 \wedge \cdots \wedge m_k.$$

Remark: The exterior algebra also satisfies a universal property similar to (3) of Theorem 34, namely with respect to R -module homomorphisms from M to R -algebras A satisfying $a^2 = 0$ for all $a \in A$ (cf. Exercise 6).

Proof: The k -tensors $\mathcal{A}^k(M)$ in the ideal $\mathcal{A}(M)$ are finite sums of elements of the form

$$m_1 \otimes \cdots \otimes m_{i-1} \otimes (m \otimes m) \otimes m_{i+2} \otimes \cdots \otimes m_k$$

with $m_1, \dots, m_k, m \in M$ (where $k \geq 2$ and $1 \leq i < k$), which is a k -tensor with two equal entries (in positions i and $i+1$), so is of the form in (1). For the reverse inclusion, note that since

$$\begin{aligned} m' \otimes m &= -m \otimes m' + [(m + m') \otimes (m + m') - m \otimes m - m' \otimes m'] \\ &\equiv -m \otimes m' \pmod{\mathcal{A}(M)}, \end{aligned}$$

interchanging any two consecutive entries and multiplying by -1 in a simple k -tensor gives an equivalent tensor modulo $\mathcal{A}^k(M)$. Using such a sequence of interchanges and sign changes we can arrange for the equal entries m_i and m_j of a simple tensor as in (1) to be adjacent, which gives an element of $\mathcal{A}^k(M)$. It follows that the generators in (1) are contained in $\mathcal{A}^k(M)$, which proves the first part of the theorem.

As in Theorem 34, the proof of (2) follows easily from the corresponding result for the tensor algebra in Theorem 31 since $\mathcal{A}^k(M)$ is contained in the kernel of any alternating map from $\mathcal{T}^k(M)$ to N .

Examples

- (1) Suppose V is a one-dimensional vector space over F with basis element v . Then $\bigwedge^k(V)$ consists of finite sums of elements of the form $\alpha_1 v \wedge \alpha_2 v \wedge \cdots \wedge \alpha_k v$, i.e., $\alpha_1 \alpha_2 \cdots \alpha_k (v \wedge v \wedge \cdots \wedge v)$ for $\alpha_1, \dots, \alpha_k \in F$. Since $v \wedge v = 0$, it follows that $\bigwedge^0(V) = F$, $\bigwedge^1(V) = V$, and $\bigwedge^i(V) = 0$ for $i \geq 2$, so as a graded F -algebra we have

$$\bigwedge(V) = F \oplus V \oplus 0 \oplus 0 \oplus \dots$$

- (2) Suppose now that V is a two-dimensional vector space over F with basis v, v' . Here $\bigwedge^k(V)$ consists of finite sums of elements of the form $(\alpha_1 v + \alpha'_1 v') \wedge \cdots \wedge (\alpha_k v + \alpha'_k v')$. Such an element is a sum of elements that are simple wedge products involving only v and v' . For example, an element in $\bigwedge^2(V)$ is a sum of elements of the form

$$\begin{aligned} (av + bv') \wedge (cv + dv') &= ac(v \wedge v) + ad(v \wedge v') + bc(v' \wedge v) \\ &\quad + bd(v' \wedge v') \\ &= (ad - bc)v \wedge v'. \end{aligned}$$

It follows that $\bigwedge^i(V) = 0$ for $i \geq 3$ since then at least one of v, v' appears twice in such simple products.

We can see directly from $\bigwedge^2(V) = \mathcal{T}^2(V)/\mathcal{A}^2(V)$ that $v \wedge v' \neq 0$, as follows. The vector space $\mathcal{T}^2(V)$ is 4-dimensional with $v \otimes v, v \otimes v', v' \otimes v, v' \otimes v'$ as basis (Proposition 16). The elements $v \otimes v, v \otimes v' + v' \otimes v, v' \otimes v'$ and $v \otimes v'$ are therefore also a basis for $\mathcal{T}^2(V)$. The subspace $\mathcal{A}^2(V)$ consists of all the 2-tensors in the ideal generated by the tensors

$$(av + bv') \otimes (av + bv') = a^2(v \otimes v) + ab(v \otimes v' + v' \otimes v) + b^2(v' \otimes v'),$$

from which it is clear that $\mathcal{A}^2(V)$ is contained in the 3-dimensional subspace having $v \otimes v, v \otimes v' + v' \otimes v$, and $v' \otimes v'$ as basis. In particular, the basis element $v \otimes v'$ of $\mathcal{T}^2(V)$ is not contained in $\mathcal{A}^2(V)$, i.e., $v \wedge v' \neq 0$ in $\bigwedge^2(V)$.

It follows that $\bigwedge^0(V) = F$, $\bigwedge^1(V) = V$, $\bigwedge^2(V) = F(v \wedge v')$, and $\bigwedge^i(V) = 0$ for $i \geq 3$, so as a graded F -algebra we have

$$\bigwedge(V) = F \oplus V \oplus F(v \wedge v') \oplus 0 \oplus \dots$$

As the previous examples illustrate, unlike the tensor and symmetric algebras, for finite dimensional vector spaces the exterior algebra is finite dimensional:

Corollary 37. Let V be a finite dimensional vector space over the field F with basis $B = \{v_1, \dots, v_n\}$. Then the vectors

$$v_{i_1} \wedge v_{i_2} \wedge \cdots \wedge v_{i_k} \quad \text{for } 1 \leq i_1 < i_2 < \cdots < i_k \leq n$$

are a basis of $\bigwedge^k(V)$, and $\bigwedge^k(V) = 0$ when $k > n$ (when $k = 0$ the basis vector is the element $1 \in F$). In particular, $\dim_F(\bigwedge^k(V)) = \binom{n}{k}$.

Proof: As the proof of Theorem 36 shows, modulo $\mathcal{A}^k(M)$, the order of the terms in any simple k -tensor can be rearranged up to introducing a sign change. It follows that the k -tensors in the corollary (which have been arranged with increasing subscripts on the v_i and with no repeated entries) are generators for $\bigwedge^k(V)$. To show these vectors are linearly independent it suffices to exhibit an alternating k -multilinear function from V^k to F which is 1 on a given $v_{i_1} \wedge v_{i_2} \wedge \cdots \wedge v_{i_k}$ and zero on all other generators. Such a function f is defined on the basis of $\mathcal{T}^k(V)$ in Proposition 32 by $f(v_{j_1} \otimes v_{j_2} \otimes \cdots \otimes v_{j_k}) = \epsilon(\sigma)$ if σ is the unique permutation of $(j_1, j_2, \dots, j_k)$ into $(i_1, i_2, \dots, i_k)$, and f is zero on every basis tensor whose k -tuple of indices cannot be permuted to $(i_1, i_2, \dots, i_k)$ (where $\epsilon(\sigma)$ is the sign of σ). Note that f is zero on any basis tensor with repeated entries. The value $\epsilon(\sigma)$ ensures that when f is extended to all elements of $\mathcal{T}^k(V)$ it gives an alternating map, i.e., f factors through $\mathcal{A}^k(V)$. Hence f is the desired function. The computation of the dimension of $\bigwedge^k(V)$ (i.e., of the number of increasing sequences of k -tuples of indices) is left to the exercises.

The results in Corollary 37 are true for any *free* R -module of rank n . In particular if $M \cong R^n$ with R -module basis $m_1, \dots, m_n$ then

$$\bigwedge^n(M) = R(m_1 \wedge \cdots \wedge m_n)$$

is a free (rank 1) R -module with generator $m_1 \wedge \cdots \wedge m_n$ and

$$\bigwedge^{n+1}(M) = \bigwedge^{n+2}(M) = \cdots = 0.$$

Example

Let R be the polynomial ring $\mathbb{Z}[x, y]$ in the variables x and y . If $M = R$, then $\bigwedge^2(M) = 0$ so, for example, there are no nontrivial alternating bilinear maps on $R \times R$ by the universal property of $\bigwedge^2(R)$ with respect to such maps (Theorem 36).

Suppose now that $M = I$ is the ideal (x, y) generated by x and y in R . Then $I \wedge I \neq 0$. Perhaps the easiest way to see this is to construct a nontrivial alternating bilinear map on $I \times I$. The map

$$\varphi(ax + by, cx + dy) = (ad - bc) \bmod (x, y)$$

is a well defined alternating R -bilinear map from $I \times I$ to $\mathbb{Z} = R/I$ (cf. Exercise 7). Since $\varphi(x, y) = 1$, it follows that $x \wedge y \in \bigwedge^2(I)$ is nonzero. Unlike the situation of free modules as in the examples following Theorem 36 (where arguments involving *bases* could be used), in this case it is not at all a trivial matter to give a direct verification that $x \wedge y \neq 0$ in $\bigwedge^2(I)$.

Remark: The ideal I is an example of a rank 1 (but *not* free) R -module (the rank of a module over an integral domain is defined in Section 12.1), and this example shows that the results of Corollary 37 are not true in general if the R -module is not free over R .

Homomorphisms of Tensor Algebras

If $\varphi : M \rightarrow N$ is any R -module homomorphism, then there is an induced map on the k^{th} tensor power:

$$T^k(\varphi) : m_1 \otimes m_2 \otimes \cdots \otimes m_k \longmapsto \varphi(m_1) \otimes \varphi(m_2) \otimes \cdots \otimes \varphi(m_k).$$

It follows directly that this map sends generators of each of the homogeneous components of the ideals $\mathcal{C}(M)$ and $\mathcal{A}(M)$ to themselves. Thus φ induces R -module homomorphisms on the quotients:

$$S^k(\varphi) : S^k(M) \longrightarrow S^k(N) \quad \text{and} \quad \bigwedge^k(\varphi) : \bigwedge^k(M) \longrightarrow \bigwedge^k(N).$$

Moreover, each of these three maps is a ring homomorphism (hence they are graded R -algebra homomorphisms).

Of particular interest is the case when $M = V$ is an n -dimensional vector space over the field F and $\varphi : V \rightarrow V$ is an endomorphism. In this case by Corollary 37, $\bigwedge^n(\varphi)$ maps the 1-dimensional space $\bigwedge^n(V)$ to itself. Let $v_1, \dots, v_n$ be a basis of V , so that $v_1 \wedge \cdots \wedge v_n$ is a basis of $\bigwedge^n(V)$. Then

$$\bigwedge^n(\varphi)(v_1 \wedge \cdots \wedge v_n) = \varphi(v_1) \wedge \cdots \wedge \varphi(v_n) = D(\varphi)v_1 \wedge \cdots \wedge v_n$$

for some scalar $D(\varphi) \in F$.

For any $n \times n$ matrix A over F we can define the associated endomorphism φ (with respect to the given basis $v_1, \dots, v_n$), which gives a map $D : M_{n \times n}(F) \rightarrow F$ where $D(A) = D(\varphi)$. It is easy to check that this map D satisfies the three axioms for a determinant function in Section 4. Then the uniqueness statement of Theorem 24 gives:

Proposition 38. If φ is an endomorphism on a n -dimensional vector space V , then $\bigwedge^n(\varphi)(w) = \det(\varphi)w$ for all $w \in \bigwedge^n(V)$.

Note that Proposition 38 characterizes the determinant of the endomorphism φ as a certain naturally induced *linear* map on $\bigwedge^n(V)$. The fact that the determinant arises naturally when considering alternating multilinear maps also explains the source of the map φ in the example above.

As with the tensor product, the maps $S^k(\varphi)$ and $\bigwedge^k(\varphi)$ induced from an injective map from M to N need not remain injective (so $\bigwedge^2(M)$ need not be a submodule of $\bigwedge^2(N)$ when M is a submodule of N , for example).

Example

The inclusion $\varphi : I \hookrightarrow R$ of the ideal (x, y) into the ring $R = \mathbb{Z}[x, y]$, both considered as R -modules, induces a map

$$\bigwedge^2(\varphi) : \bigwedge^2(I) \rightarrow \bigwedge^2(R).$$

Since $\bigwedge^2(R) = 0$ and $\bigwedge^2(I) \neq 0$, the map cannot be injective.

One can show that if M is an R -module *direct summand* of N , then $\mathcal{T}(M)$ (respectively, $\mathcal{S}(M)$ and $\bigwedge(M)$) is an R -subalgebra of $\mathcal{T}(N)$ (respectively, $\mathcal{S}(N)$ and $\bigwedge(N)$) (cf. the exercises). When $R = F$ is a field then *every* subspace M of N is a direct summand of N and so the corresponding algebra for M is a subalgebra of the algebra for N .

Symmetric and Alternating Tensors

The symmetric and exterior algebras can in some instances also be defined in terms of *symmetric* and *alternating* tensors (defined below), which identify these algebras as *subalgebras* of the tensor algebra rather than as quotient algebras.

For any R -module M there is a natural left group action of the symmetric group S_k on $M \times M \times \cdots \times M$ (k factors) given by permuting the factors:

$$\sigma(m_1, m_2, \dots, m_k) = (m_{\sigma^{-1}(1)}, m_{\sigma^{-1}(2)}, \dots, m_{\sigma^{-1}(k)}) \quad \text{for each } \sigma \in S_k$$

(the reason for σ^{-1} is to make this a *left* group action, cf. Exercise 8 of Section 5.1). This map is clearly R -multilinear, so there is a well defined R -linear left group action of S_k on $\mathcal{T}^k(M)$ which is defined on simple tensors by

$$\sigma(m_1 \otimes m_2 \otimes \cdots \otimes m_k) = m_{\sigma^{-1}(1)} \otimes m_{\sigma^{-1}(2)} \otimes \cdots \otimes m_{\sigma^{-1}(k)} \quad \text{for each } \sigma \in S_k.$$

Definition.

- (1) An element $z \in \mathcal{T}^k(M)$ is called a *symmetric k -tensor* if $\sigma z = z$ for all σ in the symmetric group S_k .
- (2) An element $z \in \mathcal{T}^k(M)$ is called an *alternating k -tensor* if $\sigma z = \epsilon(\sigma)z$ for all σ in the symmetric group S_k , where $\epsilon(\sigma)$ is the sign, ± 1 , of the permutation σ .

It is immediate from the definition that the collection of symmetric (respectively, alternating) k -tensors is an R -submodule of the module of all k -tensors.

Example

The elements $m \otimes m$ and $m_1 \otimes m_2 + m_2 \otimes m_1$ are symmetric 2-tensors. The element $m_1 \otimes m_2 - m_2 \otimes m_1$ is an alternating 2-tensor.

It is also clear from the definition that both $\mathcal{C}^k(M)$ and $\mathcal{A}^k(M)$ are stable under the action of S_k , hence there is an induced action on the quotients $\mathcal{S}^k(M)$ and $\bigwedge^k(M)$.

Proposition 39. Let σ be an element in the symmetric group S_k and let $\epsilon(\sigma)$ be the sign of the permutation σ . Then

- (1) for every $w \in \mathcal{S}^k(M)$ we have $\sigma w = w$, and
- (2) for every $w \in \bigwedge^k(M)$ we have $\sigma w = \epsilon(\sigma)w$.

Proof: The first statement is immediate from (1) in Theorem 34. We showed in the course of the proof of Theorem 36 that

$$m_1 \wedge \cdots \wedge m_i \wedge m_{i+1} \wedge \cdots \wedge m_k = -m_1 \wedge \cdots \wedge m_{i+1} \wedge m_i \wedge \cdots \wedge m_k,$$

which shows that the formula in (2) is valid on simple products for the transposition $\sigma = (i \ i+1)$. Since these transpositions generate S_k and ϵ is a group homomorphism it follows that (2) is valid for any $\sigma \in S_k$ on simple products w . Since both sides are R -linear in w , it follows that (2) holds for all $w \in \bigwedge^k(M)$.

By Proposition 39, the symmetric group S_k acts trivially on both the *submodule* of symmetric k -tensors and the *quotient* module $S^k(M)$, the k^{th} symmetric power of M . Similarly, S_k acts the same way on the submodule of alternating k -tensors as on $\bigwedge^k(M)$, the k^{th} exterior power of M . We now show that when $k!$ is a unit in R that these respective submodules and quotient modules are isomorphic (where $k!$ is the sum of the 1 of R with itself $k!$ times).

For any k -tensor $z \in \mathcal{T}^k(M)$ define

$$\begin{aligned} \text{Sym}(z) &= \sum_{\sigma \in S_k} \sigma z \\ \text{Alt}(z) &= \sum_{\sigma \in S_k} \epsilon(\sigma) \sigma z. \end{aligned}$$

For any k -tensor z , the k -tensor $\text{Sym}(z)$ is symmetric and the k -tensor $\text{Alt}(z)$ is alternating. For example, for any $\tau \in S_k$

$$\begin{aligned} \tau \text{Alt}(z) &= \sum_{\sigma \in S_k} \epsilon(\sigma) \tau \sigma z \\ &= \sum_{\sigma' \in S_k} \epsilon(\tau^{-1} \sigma') \sigma' z \quad (\text{letting } \sigma' = \tau \sigma) \\ &= \epsilon(\tau^{-1}) \sum_{\sigma' \in S_k} \epsilon(\sigma') \sigma' z = \epsilon(\tau) \text{Alt}(z). \end{aligned}$$

The tensor $\text{Sym}(z)$ is sometimes called the *symmetrization* of z and $\text{Alt}(z)$ the *skew-symmetrization* of z .

If z is already a symmetric (respectively, alternating) tensor then $\text{Sym}(z)$ (respectively, $\text{Alt}(z)$) is just $k!z$. It follows that Sym (respectively, Alt) is an R -module endomorphism of $\mathcal{T}^k(M)$ whose image lies in the submodule of symmetric (respectively, alternating) tensors. In general these maps are not surjective, but if $k!$ is a unit in R then

$$\begin{aligned} \frac{1}{k!} \text{Sym}(z) &= z \quad \text{for any symmetric tensor } z, \text{ and} \\ \frac{1}{k!} \text{Alt}(z) &= z \quad \text{for any alternating tensor } z \end{aligned}$$

so that in this case the maps $(1/k!)\text{Sym}$ and $(1/k!)\text{Alt}$ give surjective R -module homomorphisms from $\mathcal{T}^k(M)$ to the submodule of symmetric (respectively, alternating) tensors.

Proposition 40. Suppose $k!$ is a unit in the ring R and M is an R -module. Then

- (1) The map $(1/k!)Sym$ induces an R -module isomorphism between the k^{th} symmetric power of M and the R -submodule of symmetric k -tensors:

$$\frac{1}{k!}Sym : \mathcal{S}^k(M) \cong \{\text{symmetric } k\text{-tensors}\}.$$

- (2) The map $(1/k!)Alt$ induces an R -module isomorphism between the k^{th} exterior power of M and the R -submodule of alternating k -tensors:

$$\frac{1}{k!}Alt : \bigwedge^k(M) \cong \{\text{alternating } k\text{-tensors}\}.$$

Proof: We have seen that the respective maps are surjective R -homomorphisms from $\mathcal{T}^k(M)$ so to prove the proposition it suffices to check that their kernels are $\mathcal{C}^k(M)$ and $\mathcal{A}^k(M)$, respectively. We show the first and leave the second to the exercises. It is clear that Sym is 0 on any difference of two k -tensors which differ only in the order of their factors, so $\mathcal{C}^k(M)$ is contained in the kernel of $(1/k!)Sym$ by (1) of Theorem 34. For the reverse inclusion, observe that

$$z - \frac{1}{k!}Sym(z) = \frac{1}{k!} \sum_{\sigma \in S_k} (z - \sigma z)$$

for any k -tensor z . If z is in the kernel of Sym then the left hand side of this equality is just z ; and since $z - \sigma z \in \mathcal{C}^k(M)$ for every $\sigma \in S_k$ (again by (1) of Theorem 34), it follows that $z \in \mathcal{C}^k(M)$, completing the proof.

The maps $(1/k!)Sym$ and $(1/k!)Alt$ are *projections* (cf. Exercise 11 in Section 2) onto the submodules of symmetric and antisymmetric tensors, respectively. Equivalently, if $k!$ is a unit in R , we have R -module direct sums

$$\mathcal{T}^k(M) = \ker(\pi) \oplus \text{image}(\pi)$$

for $\pi = (1/k!)Sym$ or $\pi = (1/k!)Alt$. In the former case the kernel consists of $\mathcal{C}^k(M)$ and the image is the collection of symmetric tensors (in which case $\mathcal{C}^k(M)$ is said to form an R -module *complement* to the symmetric tensors). In the latter case the kernel is $\mathcal{A}^k(M)$ and the image consists of the alternating tensors.

The R -linear left group action of S_k on $\mathcal{T}^k(M)$ makes $\mathcal{T}^k(M)$ into a module over the group ring RS_k (analogous to the formation of $F[x]$ -modules described in Section 10.1). In terms of this module structure these projections give RS_k -submodule complements to the RS_k -submodules $\mathcal{C}^k(M)$ and $\mathcal{A}^k(M)$. The “averaging” technique used to construct these maps can be used to prove a very general result (Maschke’s Theorem in Section 18.1) related to actions of finite groups on vector spaces (which is the subject of the “representation theory” of finite groups in Part VI).

If $k!$ is not invertible in R then in general we do not have such S_k -invariant direct sum decompositions so it is not in general possible to identify, for example, the k^{th} exterior power of M with the alternating k -tensors of M .

Note also that when $k!$ is invertible it is possible to *define* the k^{th} exterior power of M as the collection of alternating k -tensors (this equivalent approach is sometimes found

in the literature when the theory is developed over fields such as $\mathbb{R}$ and $\mathbb{C}$). In this case the multiplication of two alternating tensors z and w is defined by first taking the product $zw = z \otimes w$ in $\mathcal{T}(M)$ and then projecting the resulting tensor into the submodule of alternating tensors. Note that the simple product of two alternating tensors need not be alternating (for example, the square of an alternating tensor is a symmetric tensor).

Example

Let V be a vector space over a field F in which $k! \neq 0$. There are many *vector space complements* to $\mathcal{A}^k(V)$ in $\mathcal{T}^k(V)$ (just extend a basis for the subspace $\mathcal{A}^k(V)$ to a basis for $\mathcal{T}^k(V)$, for example). These complements depend on choices of bases for $\mathcal{T}^k(V)$ and so are indistinguishable from each other from vector space considerations alone. The additional structure on $\mathcal{T}^k(V)$ given by the action of S_k singles out a unique complement to $\mathcal{A}^k(V)$, namely the subspace of alternating tensors in Proposition 40.

Suppose that $k! \neq 0$ in F for all $k \geq 2$ (i.e., the field F has “characteristic 0,” cf. Exercise 26 in Section 7.3), for example, $F = \mathbb{Q}$. Then the full exterior algebra $\bigwedge(V) = \bigoplus_{k \geq 0} \mathcal{A}^k(V)$ can be identified with the collection of tensors whose homogeneous components are alternating (with respect to the appropriate symmetric groups S_k).

Multiplication in $\bigwedge(V)$ in terms of alternating tensors is rather cumbersome, however. For example let v_1, v_2, v_3 be distinct basis vectors in V . The product of the two alternating tensors $z = v_1$ and $w = v_2 \otimes v_3 - v_3 \otimes v_2$ is obtained by first computing

$$z \otimes w = v_1 \otimes v_2 \otimes v_3 - v_1 \otimes v_3 \otimes v_2$$

in the full tensor algebra. This 3-tensor is not alternating — for example,

$$(1\ 2)(z \otimes w) = v_2 \otimes v_1 \otimes v_3 - v_3 \otimes v_1 \otimes v_2 \neq -z \otimes w$$

and also $(1\ 2\ 3)(z \otimes w) = v_3 \otimes v_1 \otimes v_2 - v_2 \otimes v_1 \otimes v_3 \neq z \otimes w$. The multiplication requires that we project this tensor into the subspace of alternating tensors. This projection is given by $(1/3!) \text{Alt}(z \otimes w)$ and an easy computation shows that

$$\begin{aligned} \frac{1}{6} \text{Alt}(z \otimes w) &= \frac{1}{3} [v_1 \otimes v_2 \otimes v_3 + v_2 \otimes v_3 \otimes v_1 + v_3 \otimes v_1 \otimes v_2 \\ &\quad - v_1 \otimes v_3 \otimes v_2 - v_2 \otimes v_1 \otimes v_3 - v_3 \otimes v_2 \otimes v_1], \end{aligned}$$

so the right hand side is the product of z and w in terms of alternating tensors. The same product in terms of the quotient algebra $\bigwedge(V)$ is simply

$$v_1 \wedge (2v_2 \wedge v_3) = 2v_1 \wedge v_2 \wedge v_3.$$

EXERCISES

In these exercises R is a commutative ring with 1 and M is an R -module; F is a field and V is a finite dimensional vector space over F .

1. Prove that if M is a cyclic R -module then $\mathcal{T}(M) = \mathcal{S}(M)$, i.e., the tensor algebra $\mathcal{T}(M)$ is commutative.
2. Fill in the details for the proof of Proposition 33 that $S/I = \bigoplus_{k=0}^{\infty} S_k/I_k$. [Show first that $S_i I_j \subseteq I_{i+j}$. Use this to show that the multiplication $(S_i/I_i)(S_j/I_j) \subseteq S_{i+j}/I_{i+j}$ is well defined, and then check the ring axioms and verify the statements made in the proof of Proposition 33.]

3. Show that the image of the map Sym_2 for the $\mathbb{Z}$ -module $\mathbb{Z}$ consists of the 2-tensors $a(1 \otimes 1)$ where a is an even integer. Conclude in particular that the symmetric tensor $1 \otimes 1$ in $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}$ is not contained in the image of the map Sym .
4. Prove that $m \wedge n_1 \wedge n_2 \wedge \cdots \wedge n_k = (-1)^k(n_1 \wedge n_2 \wedge \cdots \wedge n_k \wedge m)$. In particular, $x \wedge (y \wedge z) = (y \wedge z) \wedge x$ for all $x, y, z \in M$.
5. Prove that if M is a free R -module of rank n then $\bigwedge^i(M)$ is a free R -module of rank $\binom{n}{i}$ for $i = 0, 1, 2, \dots$.
6. If A is any R -algebra in which $a^2 = 0$ for all $a \in A$ and $\varphi : M \rightarrow A$ is an R -module homomorphism, prove there is a unique R -algebra homomorphism $\Phi : \bigwedge(M) \rightarrow A$ such that $\Phi|_M = \varphi$.
7. Let $R = \mathbb{Z}[x, y]$ and $I = (x, y)$.
 - (a) Prove that if $ax + by = a'x + b'y$ in R then $a' = a + yf$ and $b' = b - xf$ for some polynomial $f(x, y) \in R$.
 - (b) Prove that the map $\varphi(ax + by, cx + dy) = ad - bc \pmod{(x, y)}$ in the example following Corollary 37 is a well defined alternating R -bilinear map from $I \times I$ to $\mathbb{Z} = R/I$.
8. Let R be an integral domain and let F be its field of fractions.
 - (a) Considering F as an R -module, prove that $\bigwedge^2 F = 0$.
 - (b) Let I be any R -submodule of F (for example, any ideal in R). Prove that $\bigwedge^i I$ is a torsion R -module for $i \geq 2$ (i.e., for every $x \in \bigwedge^i I$ there is some nonzero $r \in R$ with $rx = 0$).
 - (c) Give an example of an integral domain R and an R -module I in F with $\bigwedge^i I \neq 0$ for every $i \geq 0$ (cf. the example following Corollary 37).
9. Let $R = \mathbb{Z}[G]$ be the group ring of the group $G = \{1, \sigma\}$ of order 2. Let $M = \mathbb{Z}e_1 + \mathbb{Z}e_2$ be a free $\mathbb{Z}$ -module of rank 2 with basis e_1 and e_2 . Define $\sigma(e_1) = e_1 + 2e_2$ and $\sigma(e_2) = -e_2$. Prove that this makes M into an R -module and that the R -module $\bigwedge^2 M$ is a group of order 2 with $e_1 \wedge e_2$ as generator.
10. Prove that $z - (1/k!) \text{Alt}(z) = (1/k!) \sum_{\sigma \in S_k} (z - \epsilon(\sigma)\sigma z)$ for any k -tensor z and use this to prove that the kernel of the R -module homomorphism $(1/k!) \text{Alt}$ in Proposition 40 is $\mathcal{A}^k(M)$.
11. Prove that the image of Alt_k is the unique largest subspace of $\mathcal{T}^k(V)$ on which each permutation σ in the symmetric group S_k acts as multiplication by the scalar $\epsilon(\sigma)$.
12. (a) Prove that if $f(x, y)$ is an alternating bilinear map on V (i.e., $f(x, x) = 0$ for all $x \in V$) then $f(x, y) = -f(y, x)$ for all $x, y \in V$.
 (b) Suppose that $-1 \neq 1$ in F . Prove that $f(x, y)$ is an alternating bilinear map on V (i.e., $f(x, x) = 0$ for all $x \in V$) if and only if $f(x, y) = -f(y, x)$ for all $x, y \in V$.
 (c) Suppose that $-1 = 1$ in F . Prove that every alternating bilinear form $f(x, y)$ on V is symmetric (i.e., $f(x, y) = f(y, x)$ for all $x, y \in V$). Prove that there is a symmetric bilinear map on V that is not alternating. [One approach: show that $C^2(V) \subset \mathcal{A}^2(V)$ and $C^2(V) \neq \mathcal{A}^2(V)$ by counting dimensions. Alternatively, construct an explicit symmetric map that is not alternating.]
13. Let F be any field in which $-1 \neq 1$ and let V be a vector space over F . Prove that $V \otimes_F V = \mathcal{S}^2(V) \oplus \bigwedge^2(V)$ i.e., that every 2-tensor may be written uniquely as a sum of a symmetric and an alternating tensor.
14. Prove that if M is an R -module *direct factor* of the R -module N then $\mathcal{T}(M)$ (respectively, $\mathcal{S}(M)$ and $\bigwedge(M)$) is an R -subalgebra of $\mathcal{T}(N)$ (respectively, $\mathcal{S}(N)$ and $\bigwedge(N)$).

Modules over Principal Ideal Domains

The main purpose of this chapter is to prove a structure theorem for finitely generated modules over particularly nice rings, namely Principal Ideal Domains. This theorem is an example of the ideal structure of the ring (which is particularly simple for P.I.D.s) being reflected in the structure of its modules. If we apply this result in the case where the P.I.D. is the ring of integers $\mathbb{Z}$ then we obtain a proof of the Fundamental Theorem of Finitely Generated Abelian Groups (which we examined in Chapter 5 without proof). If instead we apply this structure theorem in the case where the P.I.D. is the ring $F[x]$ of polynomials in x with coefficients in a field F we shall obtain the basic results on the so-called rational and Jordan canonical forms for a matrix. Before proceeding to the proof we briefly discuss these two important applications.

We have already discussed in Chapter 5 the result that any finitely generated abelian group is isomorphic to the direct sum of cyclic abelian groups, either $\mathbb{Z}$ or $\mathbb{Z}/n\mathbb{Z}$ for some positive integer $n \neq 0$. Recall also that an abelian group is the same thing as a $\mathbb{Z}$ -module. Since the ideals of $\mathbb{Z}$ are precisely the trivial ideal (0) and the principal ideals $(n) = n\mathbb{Z}$ generated by positive integers n , we see that the Fundamental Theorem of Finitely Generated Abelian Groups in the language of modules says that any finitely generated $\mathbb{Z}$ -module is the direct sum of modules of the form $\mathbb{Z}/I$ where I is an ideal of $\mathbb{Z}$ (these are the cyclic $\mathbb{Z}$ -modules), together with a uniqueness statement when the direct sum is written in a particular form. Note the correspondence between the ideal structure of $\mathbb{Z}$ and the structure of its (finitely generated) modules, the finitely generated abelian groups.

The Fundamental Theorem of Finitely Generated Modules over a P.I.D. states that the same result holds when the Principal Ideal Domain $\mathbb{Z}$ is replaced by *any* P.I.D. In particular, we have seen in Chapter 10 that a module over the ring $F[x]$ of polynomials in x with coefficients in the field F is the same thing as a vector space V together with a fixed linear transformation T of V (where the element x acts on V by the linear transformation T). The Fundamental Theorem in this case will say that such a vector space is the direct sum of modules of the form $F[x]/I$ where I is an ideal of $F[x]$, hence is either the trivial ideal (0) or a principal ideal $(f(x))$ generated by some nonzero polynomial $f(x)$ (these are the cyclic $F[x]$ -modules), again with a uniqueness statement when the direct sum is written in a particular form. If this is translated back into the language of vector spaces and linear transformations we can obtain information on the

linear transformation T .

For example, suppose V is a vector space of dimension n over F and we choose a basis for V . Then giving a linear transformation T of V to itself is the same thing as giving an $n \times n$ matrix A with coefficients in F (and choosing a different basis for V gives a different matrix B for T which is similar to A i.e., is of the form $P^{-1}AP$ for some invertible matrix P which defines the change of basis). We shall see that the Fundamental Theorem in this situation implies (under the assumption that the field F contains all the “eigenvalues” for the given linear transformation T) that there is a basis for V so that the associated matrix for T is *as close to being a diagonal matrix as possible* and so has a particularly simple form. This is the *Jordan canonical form*. The *rational canonical form* is another simple form for the matrix for T (that does not require the eigenvalues for T to be elements of F). In this way we shall be able to give canonical forms for arbitrary $n \times n$ matrices over fields F , that is, find matrices which are similar to a given $n \times n$ matrix and which are particularly simple (almost diagonal, for example).

Example

Let $V = \mathbb{Q}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{Q}\}$ be the usual 3-dimensional vector space of ordered 3-tuples with entries from the field $F = \mathbb{Q}$ of rational numbers and suppose T is the linear transformation

$$T(x, y, z) = (9x + 4y + 5z, -4x - 3z, -6x - 4y - 2z), \quad x, y, z \in \mathbb{Q}.$$

If we take the standard basis $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$ for V then the matrix A representing this linear transformation is

$$A = \begin{pmatrix} 9 & 4 & 5 \\ -4 & 0 & -3 \\ -6 & -4 & -2 \end{pmatrix}.$$

We shall see that the Jordan canonical form for this matrix A is the much simpler matrix

$$B = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

obtained by taking instead the basis $f_1 = (2, -1, -2)$, $f_2 = (1, 0, -1)$, $f_3 = (3, -2, -2)$ for V , since in this case

$$T(f_1) = T(2, -1, -2) = (4, -2, -4) = 2 \cdot f_1 + 0 \cdot f_2 + 0 \cdot f_3$$

$$T(f_2) = T(1, 0, -1) = (4, -1, -4) = 1 \cdot f_1 + 2 \cdot f_2 + 0 \cdot f_3$$

$$T(f_3) = T(3, -2, -2) = (9, -6, -6) = 0 \cdot f_1 + 0 \cdot f_2 + 3 \cdot f_3,$$

so the columns of the matrix representing T with respect to this basis are $(2, 0, 0)$, $(1, 2, 0)$ and $(0, 0, 3)$, i.e., T has matrix B with respect to this basis. In particular A is similar to the simpler matrix B .

In fact this linear transformation T *cannot* be diagonalized (i.e., there is no choice of basis for V for which the corresponding matrix is a diagonal matrix) so that the matrix B is as close to a diagonal matrix for T as is possible.

The first section below gives some general definitions and states and proves the Fundamental Theorem over an arbitrary P.I.D., after which we return to the application to canonical forms (the application to abelian groups appears in Chapter 5). These applications can be read independently of the general proof. An alternate and computationally useful proof valid for Euclidean Domains (so in particular for the rings $\mathbb{Z}$ and $F[x]$) along the lines of row and column operations is outlined in the exercises.

12.1 THE BASIC THEORY

We first describe some general finiteness conditions. Let R be a ring and let M be a left R -module.

Definition.

- (1) The left R -module M is said to be a *Noetherian R -module* or to satisfy the *ascending chain condition on submodules* (or *A.C.C. on submodules*) if there are no infinite increasing chains of submodules, i.e., whenever

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$$

is an increasing chain of submodules of M , then there is a positive integer m such that for all $k \geq m$, $M_k = M_m$ (so the chain becomes stationary at stage m : $M_m = M_{m+1} = M_{m+2} = \cdots$).

- (2) The ring R is said to be *Noetherian* if it is Noetherian as a left module over itself, i.e., if there are no infinite increasing chains of left ideals in R .

One can formulate analogous notions of A.C.C. on right and on two-sided ideals in a (possibly noncommutative) ring R . For noncommutative rings these properties need not be related.

Theorem 1. Let R be a ring and let M be a left R -module. Then the following are equivalent:

- (1) M is a Noetherian R -module.
- (2) Every nonempty set of submodules of M contains a maximal element under inclusion.
- (3) Every submodule of M is finitely generated.

Proof: [(1) implies (2)] Assume M is Noetherian and let Σ be any nonempty collection of submodules of M . Choose any $M_1 \in \Sigma$. If M_1 is a maximal element of Σ , (2) holds, so assume M_1 is not maximal. Then there is some $M_2 \in \Sigma$ such that $M_1 \subset M_2$. If M_2 is maximal in Σ , (2) holds, so we may assume there is an $M_3 \in \Sigma$ properly containing M_2 . Proceeding in this way one sees that if (2) fails we can produce by the Axiom of Choice an infinite strictly increasing chain of elements of Σ , contrary to (1).

[(2) implies (3)] Assume (2) holds and let N be any submodule of M . Let Σ be the collection of all finitely generated submodules of N . Since $\{0\} \in \Sigma$, this collection is nonempty. By (2) Σ contains a maximal element N' . If $N' \neq N$, let $x \in N - N'$. Since $N' \in \Sigma$, the submodule N' is finitely generated by assumption, hence also the

submodule generated by N' and x is finitely generated. This contradicts the maximality of N' , so $N = N'$ is finitely generated.

[(3) implies (1)] Assume (3) holds and let $M_1 \subseteq M_2 \subseteq M_3 \dots$ be a chain of submodules of M . Let

$$N = \bigcup_{i=1}^{\infty} M_i$$

and note that N is a submodule. By (3) N is finitely generated by, say, $a_1, a_2, \dots, a_n$. Since $a_i \in N$ for all i , each a_i lies in one of the submodules in the chain, say M_{j_i} . Let $m = \max \{j_1, j_2, \dots, j_n\}$. Then $a_i \in M_m$ for all i so the module they generate is contained in M_m , i.e., $N \subseteq M_m$. This implies $M_m = N = M_k$ for all $k \geq m$, which proves (1).

Corollary 2. If R is a P.I.D. then every nonempty set of ideals of R has a maximal element and R is a Noetherian ring.

Proof: The P.I.D. R satisfies condition (3) in the theorem with $M = R$.

Recall that even if M itself is a finitely generated R -module, submodules of M need not be finitely generated, so the condition that M be a Noetherian R -module is in general stronger than the condition that M be a finitely generated R -module.

We require a result on “linear dependence” before turning to the main results of this chapter.

Proposition 3. Let R be an integral domain and let M be a free R -module of rank $n < \infty$. Then any $n + 1$ elements of M are R -linearly dependent, i.e., for any $y_1, y_2, \dots, y_{n+1} \in M$ there are elements $r_1, r_2, \dots, r_{n+1} \in R$, not all zero, such that

$$r_1 y_1 + r_2 y_2 + \dots + r_{n+1} y_{n+1} = 0.$$

Proof: The quickest way of proving this is to embed R in its quotient field F (since R is an integral domain) and observe that since $M \cong R \oplus R \oplus \dots \oplus R$ (n times) we obtain $M \subseteq F \oplus F \oplus \dots \oplus F$. The latter is an n -dimensional vector space over F so any $n + 1$ elements of M are F -linearly dependent. By clearing the denominators of the scalars (by multiplying through by the product of all the denominators, for example), we obtain an R -linear dependence relation among the $n + 1$ elements of M .

Alternatively, let $e_1, \dots, e_n$ be a basis of the free R -module M and let $y_1, \dots, y_{n+1}$ be any $n + 1$ elements of M . For $1 \leq i \leq n + 1$ write $y_i = a_{1i}e_1 + a_{2i}e_2 + \dots + a_{ni}e_n$ in terms of the basis $e_1, e_2, \dots, e_n$. Let A be the $(n + 1) \times (n + 1)$ matrix whose i, j entry is a_{ij} , $1 \leq i \leq n, 1 \leq j \leq n + 1$ and whose last row is zero, so certainly $\det A = 0$. Since R is an integral domain, Corollary 27 of Section 11.4 shows that the columns of A are R -linearly dependent. Any dependence relation on the columns of A gives a dependence relation on the y_i 's, completing the proof.

If R is any integral domain and M is any R -module recall that

$$\text{Tor}(M) = \{x \in M \mid rx = 0 \text{ for some nonzero } r \in R\}$$

is a submodule of M (called *the* torsion submodule of M) and if N is any submodule of $\text{Tor}(M)$, N is called *a* torsion submodule of M (so the torsion submodule of M is the union of all torsion submodules of M , i.e., is the maximal torsion submodule of M). If $\text{Tor}(M) = 0$, the module M is said to be *torsion free*.

For any submodule N of M , the *annihilator* of N is the ideal of R defined by

$$\text{Ann}(N) = \{r \in R \mid rn = 0 \text{ for all } n \in N\}.$$

Note that if N is not a torsion submodule of M then $\text{Ann}(N) = (0)$. It is easy to see that if N, L are submodules of M with $N \subseteq L$, then $\text{Ann}(L) \subseteq \text{Ann}(N)$. If R is a P.I.D. and $N \subseteq L \subseteq M$ with $\text{Ann}(N) = (a)$ and $\text{Ann}(L) = (b)$, then $a \mid b$. In particular, the annihilator of any element x of M divides the annihilator of M (this is implied by Lagrange's Theorem when $R = \mathbb{Z}$).

Definition. For any integral domain R the *rank* of an R -module M is the maximum number of R -linearly independent elements of M .

The preceding proposition states that for a free R -module M over an integral domain the rank of a submodule is bounded by the rank of M . This notion of rank agrees with previous uses of the same term. If the ring $R = F$ is a field, then the rank of an R -module M is the dimension of M as a vector space over F and any maximal set of F -linearly independent elements is a basis for M . For a general integral domain, however, an R -module M of rank n need not have a “basis,” i.e., need not be a *free* R -module even if M is torsion free, so some care is necessary with the notion of rank, particularly with respect to the torsion elements of M . Exercises 1 to 6 and 20 give an alternate characterization of the rank and provide some examples of (torsion free) R -modules (of rank 1) that are not free.

The next important result shows that if N is a submodule of a free module of finite rank over a P.I.D. then N is again a free module of finite rank and furthermore it is possible to choose generators for the two modules which are related in a simple way.

Theorem 4. Let R be a Principal Ideal Domain, let M be a free R -module of finite rank n and let N be a submodule of M . Then

- (1) N is free of rank m , $m \leq n$ and
- (2) there exists a basis $y_1, y_2, \dots, y_n$ of M so that $a_1 y_1, a_2 y_2, \dots, a_m y_m$ is a basis of N where $a_1, a_2, \dots, a_m$ are nonzero elements of R with the divisibility relations

$$a_1 \mid a_2 \mid \cdots \mid a_m.$$

Proof: The theorem is trivial for $N = \{0\}$, so assume $N \neq \{0\}$. For each R -module homomorphism φ of M into R , the image $\varphi(N)$ of N is a submodule of R , i.e., an ideal in R . Since R is a P.I.D. this ideal must be principal, say $\varphi(N) = (a_\varphi)$, for some $a_\varphi \in R$. Let

$$\Sigma = \{(a_\varphi) \mid \varphi \in \text{Hom}_R(M, R)\}$$

be the collection of the principal ideals in R obtained in this way from the R -module homomorphisms of M into R . The collection Σ is certainly nonempty since taking φ

to be the trivial homomorphism shows that $(0) \in \Sigma$. By Corollary 2, Σ has at least one maximal element i.e., there is at least one homomorphism v of M to R so that the principal ideal $v(N) = (a_v)$ is not properly contained in any other element of Σ . Let $a_1 = a_v$ for this maximal element and let $y \in N$ be an element mapping to the generator a_1 under the homomorphism v : $v(y) = a_1$.

We now show the element a_1 is nonzero. Let $x_1, x_2, \dots, x_n$ be any basis of the free module M and let $\pi_i \in \text{Hom}_R(M, R)$ be the natural projection homomorphism onto the i^{th} coordinate with respect to this basis. Since $N \neq \{0\}$, there exists an i such that $\pi_i(N) \neq 0$, which in particular shows that Σ contains more than just the trivial ideal (0) . Since (a_1) is a maximal element of Σ it follows that $a_1 \neq 0$.

We next show that this element a_1 divides $\varphi(y)$ for every $\varphi \in \text{Hom}_R(M, R)$. To see this let d be a generator for the principal ideal generated by a_1 and $\varphi(y)$. Then d is a divisor of both a_1 and $\varphi(y)$ in R and $d = r_1 a_1 + r_2 \varphi(y)$ for some $r_1, r_2 \in R$. Consider the homomorphism $\psi = r_1 v + r_2 \varphi$ from M to R . Then $\psi(y) = (r_1 v + r_2 \varphi)(y) = r_1 a_1 + r_2 \varphi(y) = d$ so that $d \in \psi(N)$, hence also $(d) \subseteq \psi(N)$. But d is a divisor of a_1 so we also have $(a_1) \subseteq (d)$. Then $(a_1) \subseteq (d) \subseteq \psi(N)$ and by the maximality of (a_1) we must have equality: $(a_1) = (d) = \psi(N)$. In particular $(a_1) = (d)$ shows that $a_1 \mid \varphi(y)$ since d divides $\varphi(y)$.

If we apply this to the projection homomorphisms π_i we see that a_1 divides $\pi_i(y)$ for all i . Write $\pi_i(y) = a_1 b_i$ for some $b_i \in R$, $1 \leq i \leq n$ and define

$$y_1 = \sum_{i=1}^n b_i x_i.$$

Note that $a_1 y_1 = y$. Since $a_1 = v(y) = v(a_1 y_1) = a_1 v(y_1)$ and a_1 is a nonzero element of the integral domain R this shows

$$v(y_1) = 1.$$

We now verify that this element y_1 can be taken as one element in a basis for M and that $a_1 y_1$ can be taken as one element in a basis for N , namely that we have

- (a) $M = Ry_1 \oplus \ker v$, and
- (b) $N = Ra_1 y_1 \oplus (N \cap \ker v)$.

To see (a) let x be an arbitrary element in M and write $x = v(x)y_1 + (x - v(x)y_1)$. Since

$$\begin{aligned} v(x - v(x)y_1) &= v(x) - v(x)v(y_1) \\ &= v(x) - v(x) \cdot 1 \\ &= 0 \end{aligned}$$

we see that $x - v(x)y_1$ is an element in the kernel of v . This shows that x can be written as the sum of an element in Ry_1 and an element in the kernel of v , so $M = Ry_1 + \ker v$. To see that the sum is direct, suppose ry_1 is also an element in the kernel of v . Then $0 = v(ry_1) = rv(y_1) = r$ shows that this element is indeed 0.

For (b) observe that $v(x')$ is divisible by a_1 for every $x' \in N$ by the definition of a_1 as a generator for $v(N)$. If we write $v(x') = ba_1$ where $b \in R$ then the decomposition we used in (a) above is $x' = v(x')y_1 + (x' - v(x')y_1) = ba_1 y_1 + (x' - ba_1 y_1)$ where the second summand is in the kernel of v and is an element of N . This shows that

$N = Ra_1y_1 + (N \cap \ker \nu)$. The fact that the sum in (b) is direct is a special case of the directness of the sum in (a).

We now prove part (1) of the theorem by induction on the rank, m , of N . If $m = 0$, then N is a torsion module, hence $N = 0$ since a free module is torsion free, so (1) holds trivially. Assume then that $m > 0$. Since the sum in (b) above is direct we see easily that $N \cap \ker \nu$ has rank $m - 1$ (cf. Exercise 3). By induction $N \cap \ker \nu$ is then a free R -module of rank $m - 1$. Again by the directness of the sum in (b) we see that adjoining a_1y_1 to any basis of $N \cap \ker \nu$ gives a basis of N , so N is also free (of rank m), which proves (1).

Finally, we prove (2) by induction on n , the rank of M . Applying (1) to the submodule $\ker \nu$ shows that this submodule is free and because the sum in (a) is direct it is free of rank $n - 1$. By the induction assumption applied to the module $\ker \nu$ (which plays the role of M) and its submodule $\ker \nu \cap N$ (which plays the role of N), we see that there is a basis $y_2, y_3, \dots, y_n$ of $\ker \nu$ such that $a_2y_2, a_3y_3, \dots, a_my_m$ is a basis of $N \cap \ker \nu$ for some elements $a_2, a_3, \dots, a_m$ of R with $a_2 \mid a_3 \mid \dots \mid a_m$. Since the sums (a) and (b) are direct, $y_1, y_2, \dots, y_n$ is a basis of M and $a_1y_1, a_2y_2, \dots, a_my_m$ is a basis of N . To complete the induction it remains to show that a_1 divides a_2 . Define a homomorphism φ from M to R by defining $\varphi(y_1) = \varphi(y_2) = 1$ and $\varphi(y_i) = 0$, for all $i > 2$, on the basis for M . Then for this homomorphism φ we have $a_1 = \varphi(a_1y_1)$ so $a_1 \in \varphi(N)$ hence also $(a_1) \subseteq \varphi(N)$. By the maximality of (a_1) in Σ it follows that $(a_1) = \varphi(N)$. Since $a_2 = \varphi(a_2y_2) \in \varphi(N)$ we then have $a_2 \in (a_1)$ i.e., $a_1 \mid a_2$. This completes the proof of the theorem.

Recall that the left R -module C is a *cyclic R -module* (for any ring R , not necessarily commutative nor with 1) if there is an element $x \in C$ such that $C = Rx$. We can then define an R -module homomorphism

$$\pi : R \rightarrow C$$

by $\pi(r) = rx$, which will be surjective by the assumption $C = Rx$. The First Isomorphism Theorem gives an isomorphism of (left) R -modules

$$R/\ker \pi \cong C.$$

If R is a P.I.D., $\ker \pi$ is a principal ideal, (a) , so we see that the cyclic R -modules C are of the form $R/(a)$ where $(a) = \text{Ann}(C)$.

The cyclic modules are the simplest modules (since they require only one generator). The existence portion of the Fundamental Theorem states that any finitely generated module over a P.I.D. is isomorphic to the direct sum of finitely many cyclic modules.

Theorem 5. (Fundamental Theorem, Existence: Invariant Factor Form) Let R be a P.I.D. and let M be a finitely generated R -module.

(1) Then M is isomorphic to the direct sum of finitely many cyclic modules. More precisely,

$$M \cong R^r \oplus R/(a_1) \oplus R/(a_2) \oplus \dots \oplus R/(a_m)$$

for some integer $r \geq 0$ and nonzero elements $a_1, a_2, \dots, a_m$ of R which are not units in R and which satisfy the divisibility relations

$$a_1 \mid a_2 \mid \dots \mid a_m.$$

- (2) M is torsion free if and only if M is free.
 (3) In the decomposition in (1),

$$\text{Tor}(M) \cong R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m).$$

In particular M is a torsion module if and only if $r = 0$ and in this case the annihilator of M is the ideal (a_m) .

Proof: The module M can be generated by a finite set of elements by assumption so let $x_1, x_2, \dots, x_n$ be a set of generators of M of minimal cardinality. Let R^n be the free R -module of rank n with basis $b_1, b_2, \dots, b_n$ and define the homomorphism $\pi : R^n \rightarrow M$ by defining $\pi(b_i) = x_i$ for all i , which is automatically surjective since $x_1, \dots, x_n$ generate M . By the First Isomorphism Theorem for modules we have $R^n / \ker \pi \cong M$. Now, by Theorem 4 applied to R^n and the submodule $\ker \pi$ we can choose another basis $y_1, y_2, \dots, y_n$ of R^n so that $a_1 y_1, a_2 y_2, \dots, a_m y_m$ is a basis of $\ker \pi$ for some elements $a_1, a_2, \dots, a_m$ of R with $a_1 \mid a_2 \mid \cdots \mid a_m$. This implies

$$M \cong R^n / \ker \pi = (Ry_1 \oplus Ry_2 \oplus \cdots \oplus Ry_n) / (Ra_1 y_1 \oplus Ra_2 y_2 \oplus \cdots \oplus Ra_m y_m).$$

To identify the quotient on the right hand side we use the natural surjective R -module homomorphism

$$Ry_1 \oplus Ry_2 \oplus \cdots \oplus Ry_n \rightarrow R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m) \oplus R^{n-m}$$

that maps $(\alpha_1 y_1, \dots, \alpha_n y_n)$ to $(\alpha_1 \bmod (a_1), \dots, \alpha_m \bmod (a_m), \alpha_{m+1}, \dots, \alpha_n)$. The kernel of this map is clearly the set of elements where a_i divides α_i , $i = 1, 2, \dots, m$, i.e., $Ra_1 y_1 \oplus Ra_2 y_2 \oplus \cdots \oplus Ra_m y_m$ (cf. Exercise 7). Hence we obtain

$$M \cong R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m) \oplus R^{n-m}.$$

If a is a unit in R then $R/(a) = 0$, so in this direct sum we may remove any of the initial a_i which are units. This gives the decomposition in (1) (with $r = n - m$).

Since $R/(a)$ is a torsion R -module for any nonzero element a of R , (1) immediately implies M is a torsion free module if and only if $M \cong R^r$, which is (2). Part (3) is immediate from the definitions since the annihilator of $R/(a)$ is evidently the ideal (a) .

We shall shortly prove the uniqueness of the decomposition in Theorem 5, namely that if we have

$$M \cong R^{r'} \oplus R/(b_1) \oplus R/(b_2) \oplus \cdots \oplus R/(b_{m'})$$

for some integer $r' \geq 0$ and nonzero elements $b_1, b_2, \dots, b_{m'}$ of R which are not units with

$$b_1 \mid b_2 \mid \cdots \mid b_{m'},$$

then $r = r'$, $m = m'$ and $(a_i) = (b_i)$ (so $a_i = b_i$ up to units) for all i . It is precisely the divisibility condition $a_1 \mid a_2 \mid \cdots \mid a_m$ which gives this uniqueness.

Definition. The integer r in Theorem 5 is called the *free rank* or the *Betti number* of M and the elements $a_1, a_2, \dots, a_m \in R$ (defined up to multiplication by units in R) are called the *invariant factors* of M .

Note that until we have proved that the invariant factors of M are unique we should properly refer to a set of invariant factors for M (and similarly for the free rank), by which we mean any elements giving a decomposition for M as in (1) of the theorem above.

Using the Chinese Remainder Theorem it is possible to decompose the cyclic modules in Theorem 5 further so that M is the direct sum of cyclic modules whose annihilators are as simple as possible (namely (0) or generated by powers of primes in R). This gives an alternate decomposition which we shall also see is unique and which we now describe.

Suppose a is a nonzero element of the Principal Ideal Domain R . Then since R is also a Unique Factorization Domain we can write

$$a = up_1^{\alpha_1} p_2^{\alpha_2} \dots p_s^{\alpha_s}$$

where the p_i are distinct primes in R and u is a unit. This factorization is unique up to units, so the ideals $(p_i^{\alpha_i})$, $i = 1, \dots, s$ are uniquely defined. For $i \neq j$ we have $(p_i^{\alpha_i}) + (p_j^{\alpha_j}) = R$ since the sum of these two ideals is generated by a greatest common divisor, which is 1 for distinct primes p_i, p_j . Put another way, the ideals $(p_i^{\alpha_i})$, $i = 1, \dots, s$, are comaximal in pairs. The intersection of all these ideals is the ideal (a) since a is the least common multiple of $p_1^{\alpha_1}, p_2^{\alpha_2}, \dots, p_s^{\alpha_s}$. Then the Chinese Remainder Theorem (Theorem 7.17) shows that

$$R/(a) \cong R/(p_1^{\alpha_1}) \oplus R/(p_2^{\alpha_2}) \oplus \dots \oplus R/(p_s^{\alpha_s})$$

as rings and also as R -modules.

Applying this to the modules in Theorem 5 allows us to write each of the direct summands $R/(a_i)$ for the invariant factor a_i of M as a direct sum of cyclic modules whose annihilators are the prime power divisors of a_i . This proves:

Theorem 6. (Fundamental Theorem, Existence: Elementary Divisor Form) Let R be a P.I.D. and let M be a finitely generated R -module. Then M is the direct sum of a finite number of cyclic modules whose annihilators are either (0) or generated by powers of primes in R , i.e.,

$$M \cong R^r \oplus R/(p_1^{\alpha_1}) \oplus R/(p_2^{\alpha_2}) \oplus \dots \oplus R/(p_t^{\alpha_t})$$

where $r \geq 0$ is an integer and $p_1^{\alpha_1}, \dots, p_t^{\alpha_t}$ are positive powers of (not necessarily distinct) primes in R .

We proved Theorem 6 by using the prime power factors of the invariant factors for M . In fact we shall see that the decomposition of M into a direct sum of cyclic modules whose annihilators are (0) or prime powers as in Theorem 6 is unique, i.e., the integer r and the ideals $(p_1^{\alpha_1}), \dots, (p_t^{\alpha_t})$ are uniquely defined for M . These prime powers are given a name:

Definition. Let R be a P.I.D. and let M be a finitely generated R -module as in Theorem 6. The prime powers $p_1^{\alpha_1}, \dots, p_i^{\alpha_i}$ (defined up to multiplication by units in R) are called the *elementary divisors* of M .

Suppose M is a finitely generated torsion module over the Principal Ideal Domain R . If for the *distinct* primes $p_1, p_2, \dots, p_n$ occurring in the decomposition in Theorem 6 we group together all the cyclic factors corresponding to the same prime p_i we see in particular that M can be written as a direct sum

$$M = N_1 \oplus N_2 \oplus \cdots \oplus N_n$$

where N_i consists of all the elements of M which are annihilated by some power of the prime p_i . This result holds also for modules over R which may not be finitely generated:

Theorem 7. (The Primary Decomposition Theorem) Let R be a P.I.D. and let M be a nonzero torsion R -module (not necessarily finitely generated) with nonzero annihilator a . Suppose the factorization of a into distinct prime powers in R is

$$a = up_1^{\alpha_1} p_2^{\alpha_2} \cdots p_n^{\alpha_n}$$

and let $N_i = \{x \in M \mid p_i^{\alpha_i} x = 0\}$, $1 \leq i \leq n$. Then N_i is a submodule of M with annihilator $p_i^{\alpha_i}$ and is the submodule of M of all elements annihilated by some power of p_i . We have

$$M = N_1 \oplus N_2 \oplus \cdots \oplus N_n.$$

If M is finitely generated then each N_i is the direct sum of finitely many cyclic modules whose annihilators are divisors of $p_i^{\alpha_i}$.

Proof: We have already proved these results in the case where M is finitely generated over R . In the general case it is clear that N_i is a submodule of M with annihilator dividing $p_i^{\alpha_i}$. Since R is a P.I.D. the ideals $(p_i^{\alpha_i})$ and $(p_j^{\alpha_j})$ are comaximal for $i \neq j$, so the direct sum decomposition of M can be proved easily by modifying the argument in the proof of the Chinese Remainder Theorem to apply it to modules. Using this direct sum decomposition it is easy to see that the annihilator of N_i is precisely $p_i^{\alpha_i}$.

Definition. The submodule N_i in the previous theorem is called the *p_i -primary component* of M .

Notice that with this terminology the elementary divisors of a finitely generated module M are just the invariant factors of the primary components of $\text{Tor}(M)$.

We now prove the uniqueness statements regarding the decompositions in the Fundamental Theorem.

Note that if M is any module over a commutative ring R and a is an element of R then $aM = \{am \mid m \in M\}$ is a submodule of M . Recall also that in a Principal Ideal Domain R the nonzero prime ideals are maximal, hence the quotient of R by a nonzero prime ideal is a field.

Lemma 8. Let R be a P.I.D. and let p be a prime in R . Let F denote the field $R/(p)$.

- (1) Let $M = R^r$. Then $M/pM \cong F^r$.
- (2) Let $M = R/(a)$ where a is a nonzero element of R . Then

$$M/pM \cong \begin{cases} F & \text{if } p \text{ divides } a \text{ in } R \\ 0 & \text{if } p \text{ does not divide } a \text{ in } R. \end{cases}$$

- (3) Let $M = R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_k)$ where each a_i is divisible by p . Then $M/pM \cong F^k$.

Proof: (1) There is a natural map from R^r to $(R/(p))^r$ defined by mapping $(\alpha_1, \dots, \alpha_r)$ to $(\alpha_1 \bmod (p), \dots, \alpha_r \bmod (p))$. This is clearly a surjective R -module homomorphism with kernel consisting of the r -tuples all of whose coordinates are divisible by p , i.e., pR^r , so $R^r/pR^r \cong (R/(p))^r$, which is (1).

(2) This follows from the Isomorphism Theorems: note first that $p(R/(a))$ is the image of the ideal (p) in the quotient $R/(a)$, hence is $(p) + (a)/(a)$. The ideal $(p) + (a)$ is generated by a greatest common divisor of p and a , hence is (p) if p divides a and is $R = (1)$ otherwise. Hence $pM = (p)/(a)$ if p divides a and is $R/(a) = M$ otherwise. If p divides a then $M/pM = (R/(a))/((p)/(a)) \cong R/(p)$, and if p does not divide a then $M/pM = M/M = 0$, which proves (2).

- (3) This follows from (2) as in the proof of part (1) of Theorem 5.

Theorem 9. (Fundamental Theorem, Uniqueness) Let R be a P.I.D.

- (1) Two finitely generated R -modules M_1 and M_2 are isomorphic if and only if they have the same free rank and the same list of invariant factors.
- (2) Two finitely generated R -modules M_1 and M_2 are isomorphic if and only if they have the same free rank and the same list of elementary divisors.

Proof: If M_1 and M_2 have the same free rank and list of invariant factors or the same free rank and list of elementary divisors then they are clearly isomorphic.

Suppose that M_1 and M_2 are isomorphic. Any isomorphism between M_1 and M_2 maps the torsion in M_1 to the torsion in M_2 so we must have $\text{Tor}(M_1) \cong \text{Tor}(M_2)$. Then $R^{r_1} \cong M_1/\text{Tor}(M_1) \cong M_2/\text{Tor}(M_2) \cong R^{r_2}$ where r_1 is the free rank of M_1 and r_2 is the free rank of M_2 . Let p be any nonzero prime in R . Then from $R^{r_1} \cong R^{r_2}$ we obtain $R^{r_1}/pR^{r_1} \cong R^{r_2}/pR^{r_2}$. By (1) of the previous lemma, this implies $F^{r_1} \cong F^{r_2}$ where F is the field R/pR . Hence we have an isomorphism of an r_1 -dimensional vector space over F with an r_2 -dimensional vector space over F , so that $r_1 = r_2$ and M_1 and M_2 have the same free rank.

We are reduced to showing that M_1 and M_2 have the same lists of invariant factors and elementary divisors. To do this we need only work with the isomorphic torsion modules $\text{Tor}(M_1)$ and $\text{Tor}(M_2)$, i.e., we may as well assume that both M_1 and M_2 are torsion R -modules.

We first show they have the same elementary divisors. It suffices to show that for any fixed prime p the elementary divisors which are a power of p are the same for both M_1 and M_2 . If $M_1 \cong M_2$ then the p -primary submodule of M_1 (= the direct